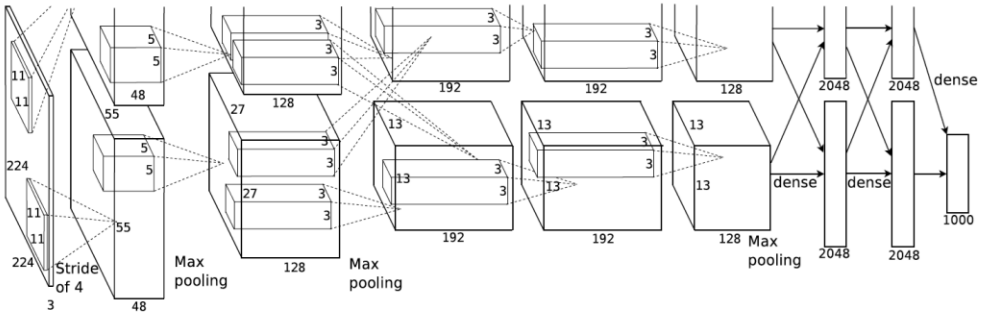
Hyperbolic Deep Learning

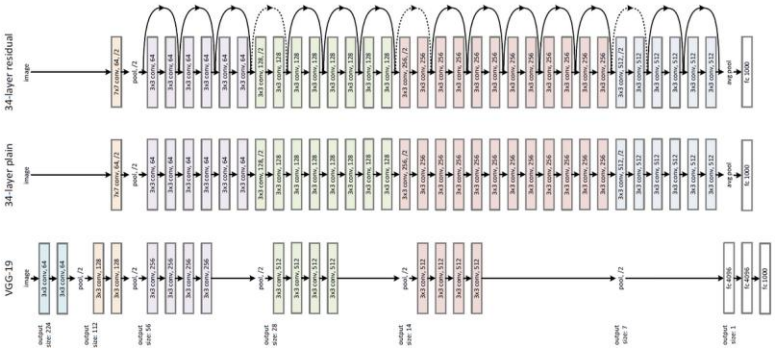
Advanced Machine Learning and Computer Vision
6 November 2026
Prof. Fabio Galasso

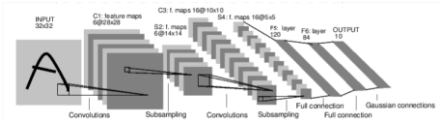


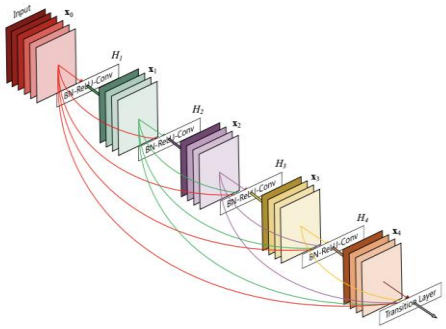
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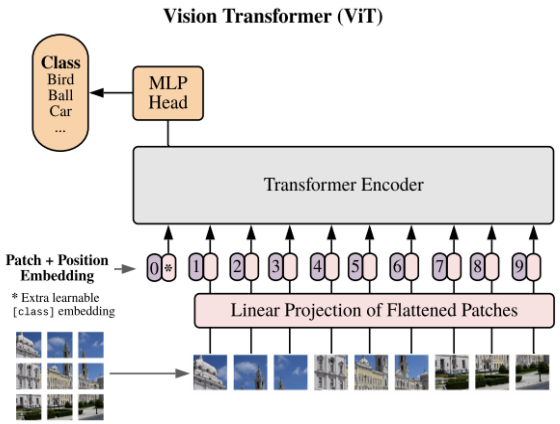
The long cycle of new deep network architectures

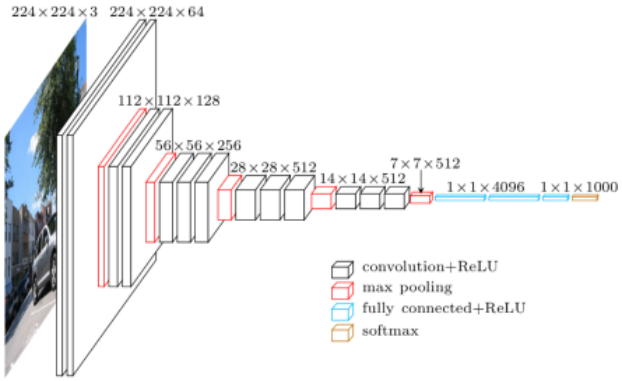






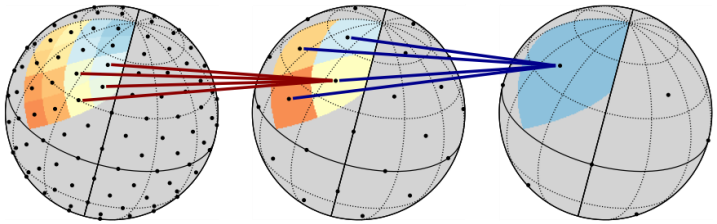




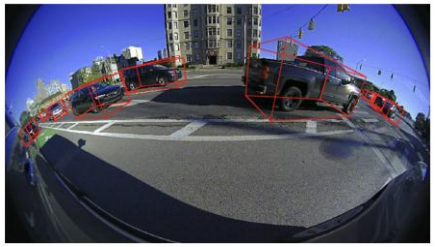


But what if data is not on regular grids?

- Often, our data is not best represented in Euclidean space



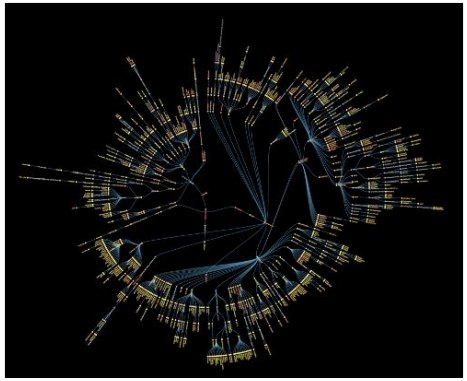
Perraudin et al. Astronomy and Computing 2019



Plaut et al. CVPRw 2021

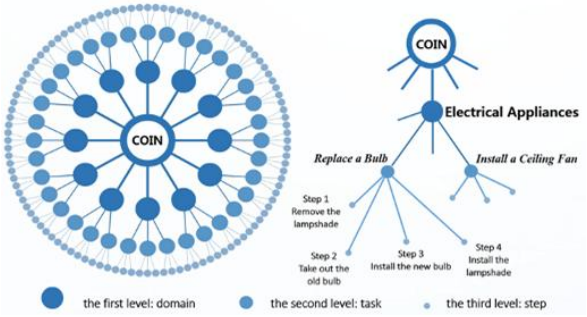


- How to do *deep learning* in such settings?



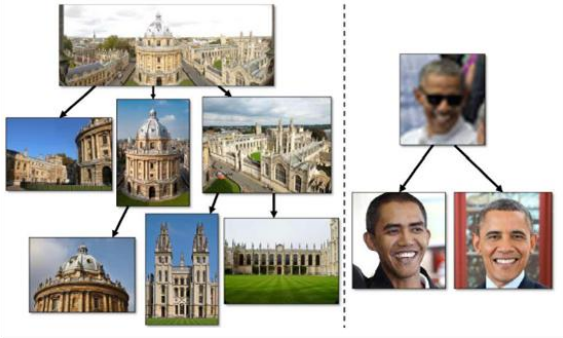
Hierarchies are everywhere!

Tang et al. CVPR 2019



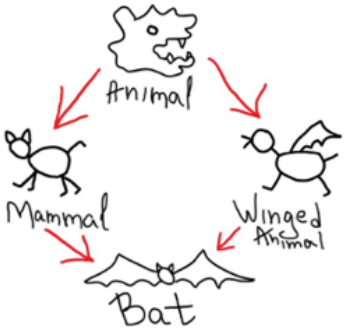
Semantics are commonly hierarchical

Khrulkov et al. CVPR 2020



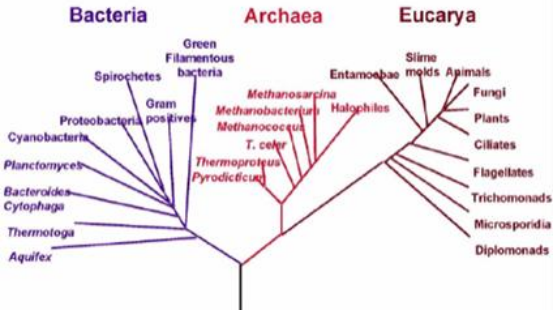
Visual collections are commonly hierarchical

Franco et al. ICLR 2023



End-to-end estimate of uncertainty

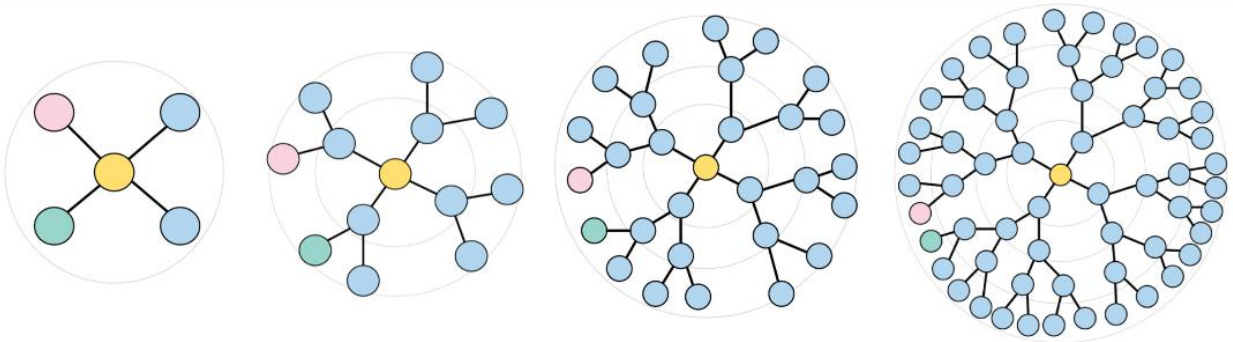
Zhang et al. Nat. Neur. 2022



Taxonomies

Hyperbolic geometry: the natural geometry of hierarchies

- Hierarchies and Euclidean space are a mismatch
  - exponential vs. linear growth



Bachmann et al. ICML 2020

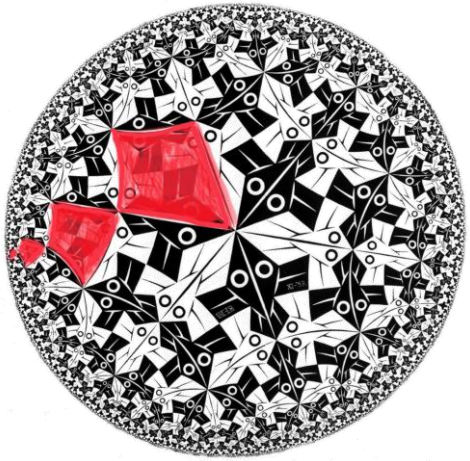


M. C. Escher (1958)

- In hyperbolic space, distances also grow exponentially
- This allows us to embed hierarchical data with minimal distortion

Issues with Euclidean Geometry

- The Euclidean space is too “narrow”



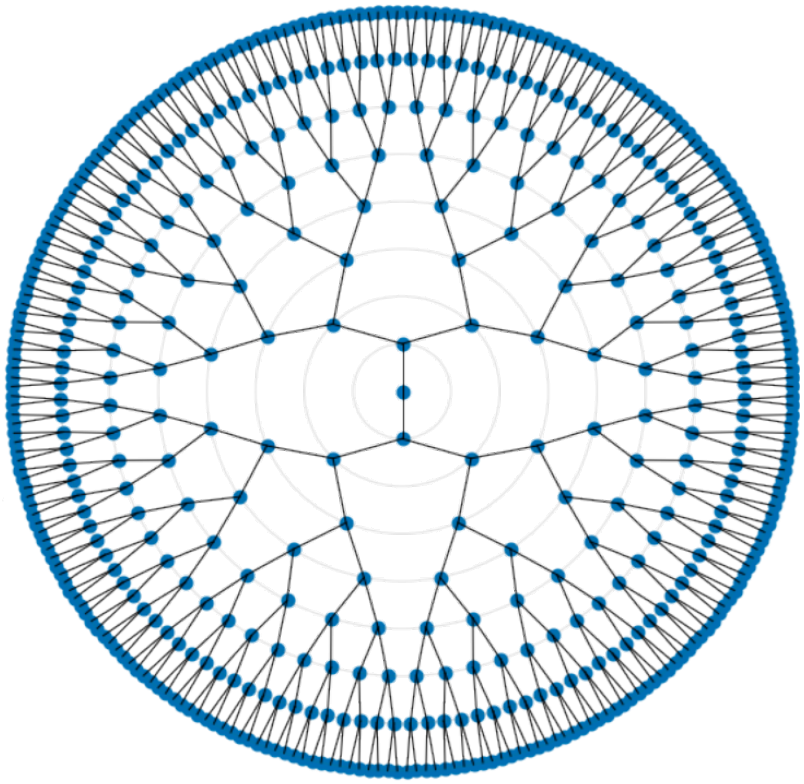
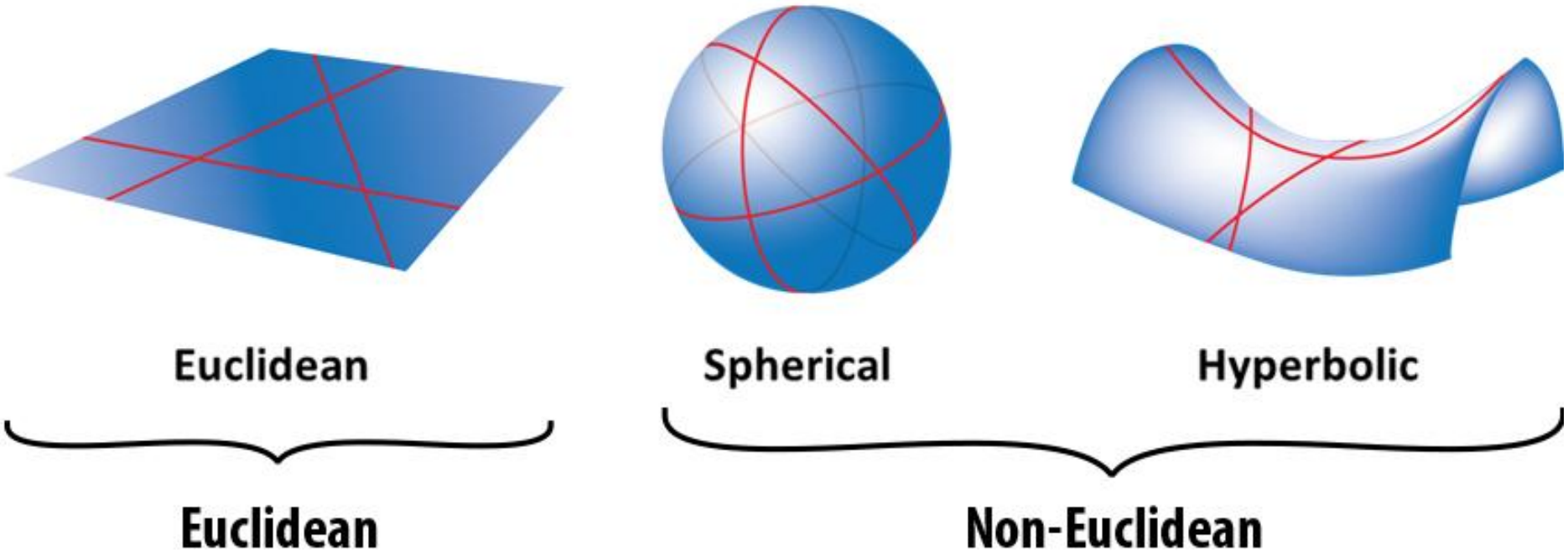


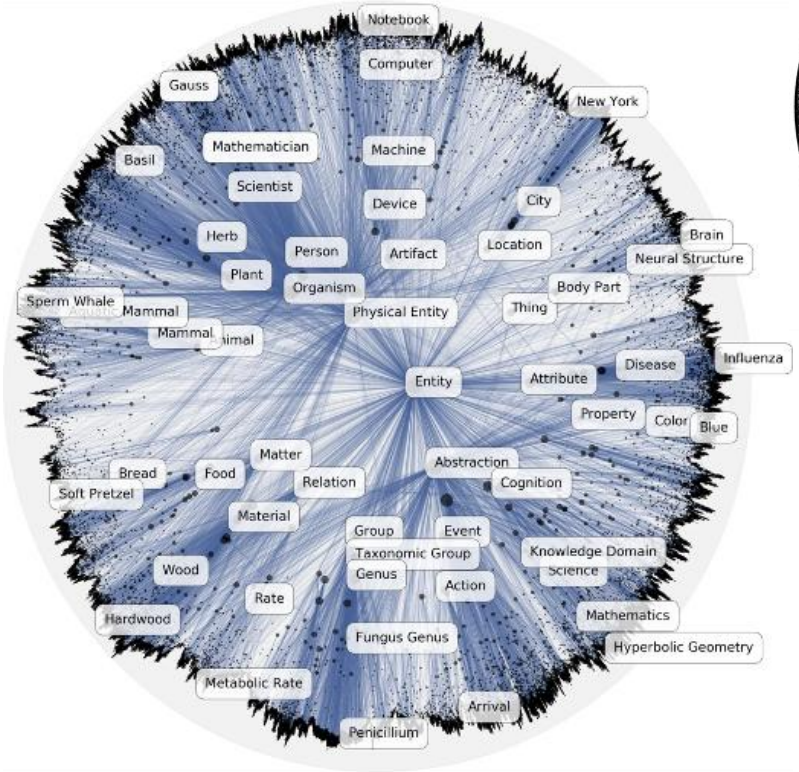
Image credit: building-babylon.net

In Euclidean space, trees are distorted [Lee, Naor, Peres’07]

Curved and non-curved spaces

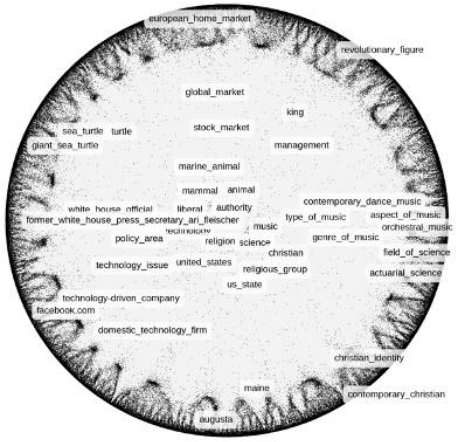


Early success: embedding hierarchies, graphs, and text



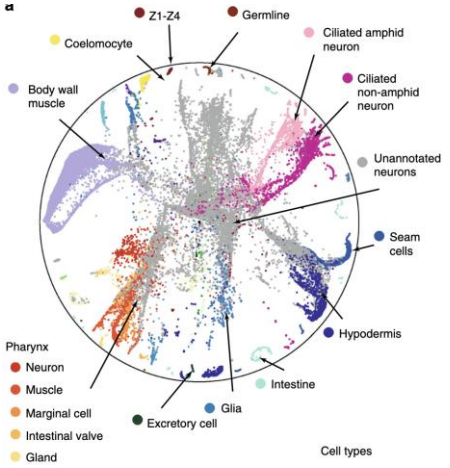
Hierarchical text embeddings

Nickel and Kiela. NeurIPS 2017



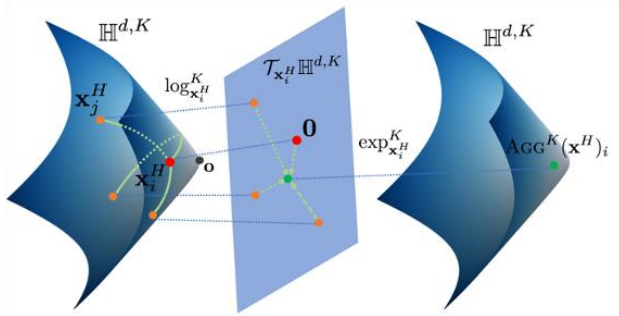
Hierarchical text embeddings

Tifrea et al. ICLR 2019



Molecular regrepresentation learning

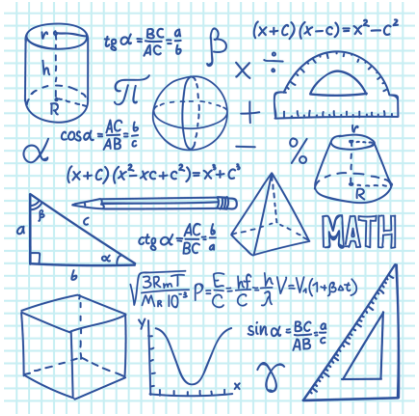
Klimovskaia et al. Nat Comm 2020



GNN on Riemannian manifold

Liu et al. NeurIPS 2019

Why is hyperbolic learning not the standard (yet)?

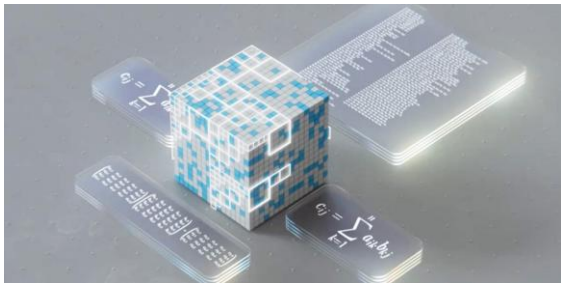


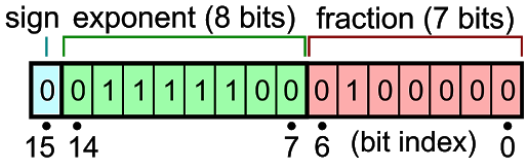
Our school curricula are Euclidean

PyTorch

TensorFlow

Our deep learning tools are Euclidean

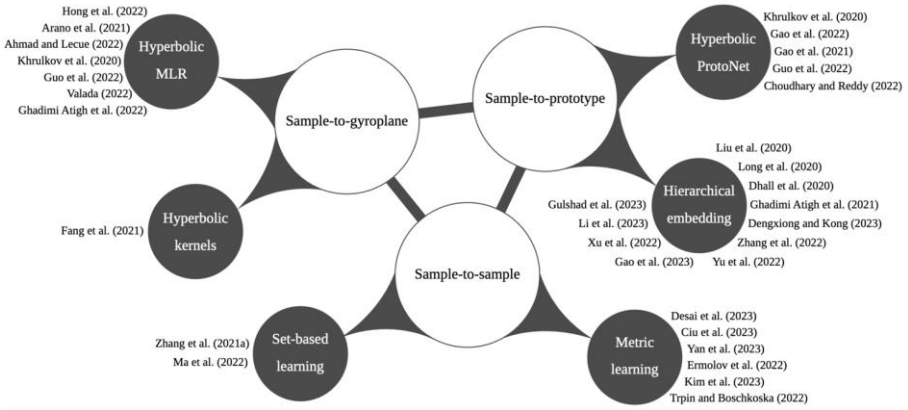




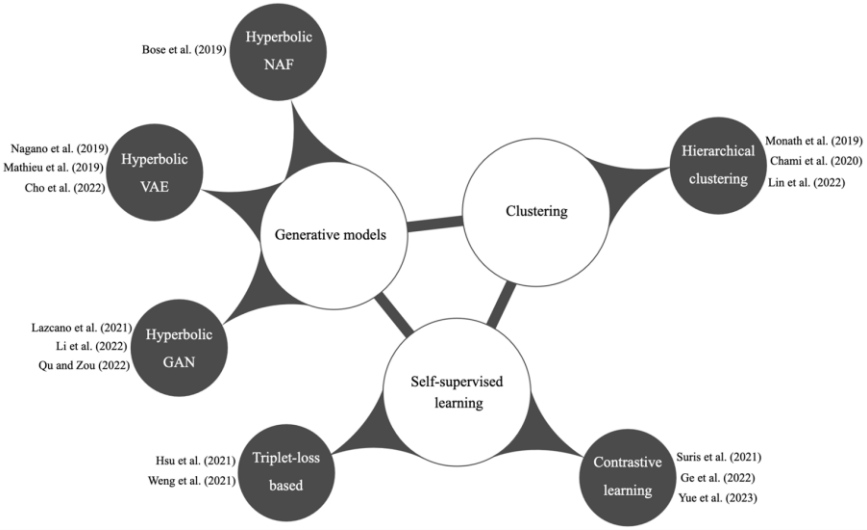
Our computers work best with Euclidean data

How far are we with hyperbolic learning in vision?

Supervised learning



Unsupervised learning



Mettes et al. Hyperbolic Deep Learning in Computer Vision: A Survey. *arXiv:2305.06611*. 2023

Overview

- What is hyperbolic geometry?
- Overview of the field
- From Euclid to Hyperbolic Deep Learning
  - Riemmanian optimizers
  - Hyperbolic radius
  - Entailment cones
- Leading Interpretations of the Hyperbolic Radius
  - Hyperbolic Uncertainty for Anomaly Detection
  - Hyperbolic Uncertainty for (Self-Paced) Self-Supervised Learning
- Open Research Perspectives on the Hyperbolic Radius
- Closing remarks

Overview

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What is hyperbolic geometry?

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Brief History of Hyperbolic Studies

300 a.C.

Euclid “Elementi”



Euclid

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Back to History: Euclidean Geometry

Five Euclid's axioms (300 b.C. "Elements"):

I. A straight line may be drawn between any two points

II. Any line segment straight line may be extended indefinitely

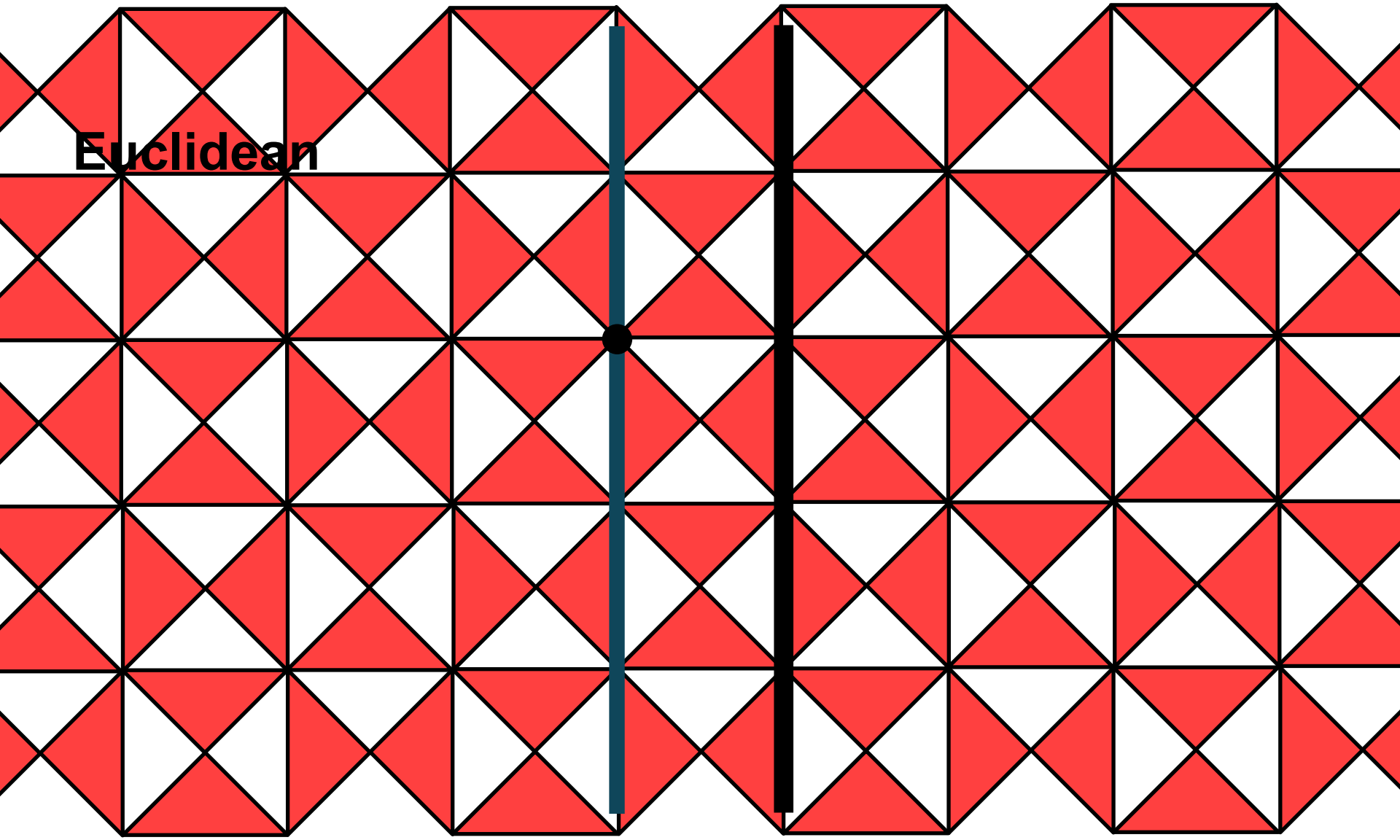
III. A circle may be drawn with any point as center and any radius

IV. All right angles are equal

V. Given a line and a point not it, at most one parallel line can be drawn



Euclid


Euclidean

Brief History of Hyperbolic Studies



Nikolay Ivanovich Lobachevsky



Carl Friedrich Gauss



János Bolyai

300 a.C. — **1829**

Euclid "Elementi" — Invention



Euclid

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Hyperbolic Geometry

Five Euclid's axioms (300 b.C. "Elements"):

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Euclid



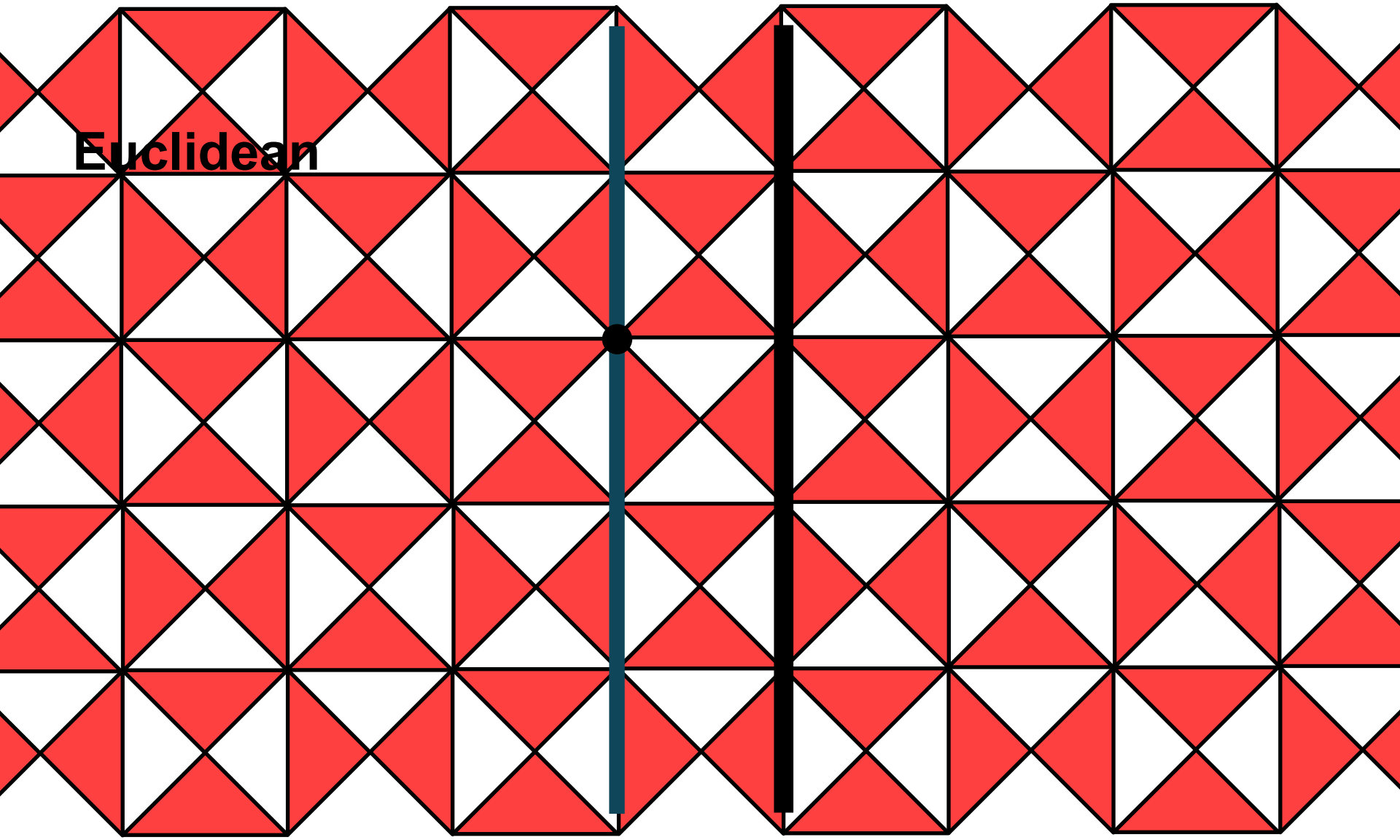
Nikolay Ivanovich **Lobachevsky**



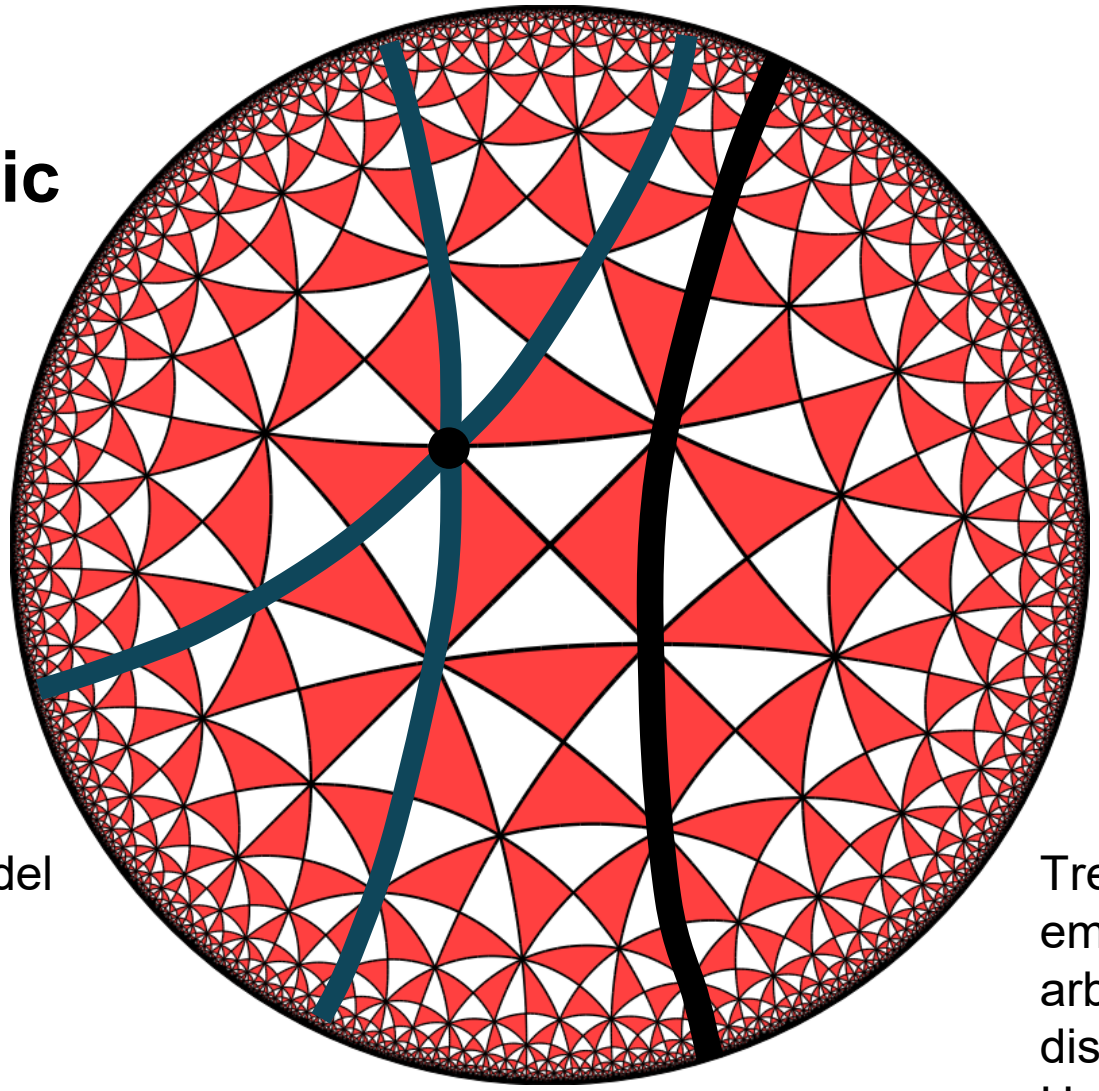
Carl Friedrich **Gauss**



János **Bolyai**


Euclidean

Hyperbolic



Poincaré ball model

Trees can be embedded with arbitrary small distortion in Hyperbolic [Sarkar GD'11]

Brief History of Hyperbolic Studies



Nikolay Ivanovich Lobachevsky



Carl Friedrich Gauss



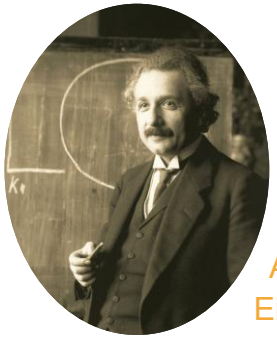
János Bolyai

300 a.C. — **1829**

Euclid "Elementi" — Invention



Euclid



Albert Einstein



Abraham A. Ungar

More on curvature

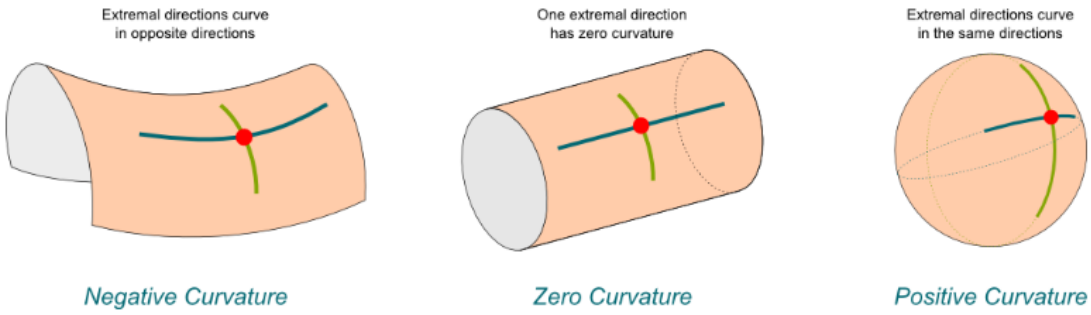
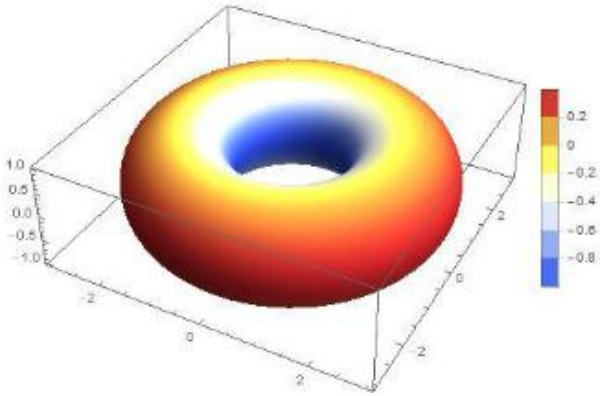


Figure 1: Examples of the three different types of Gaussian curvature (source: Science4All).



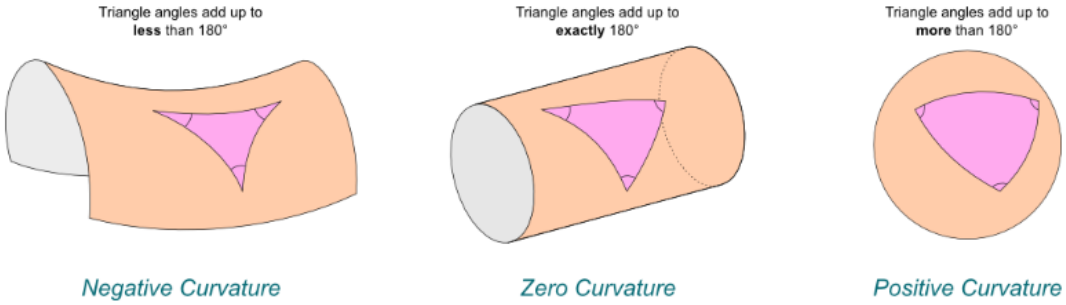


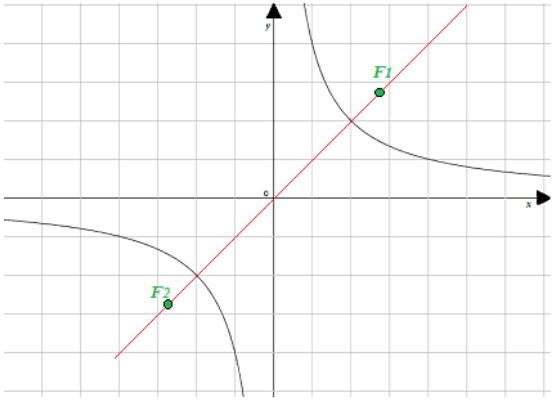
Figure 3: Deviation of a triangle under the three different types of Gaussian Curvature (source: Science4All).

The Hyperboloid model

From school we (should) remember

$$\mathcal{H}: x^2 - y^2 = -1$$

What is this?



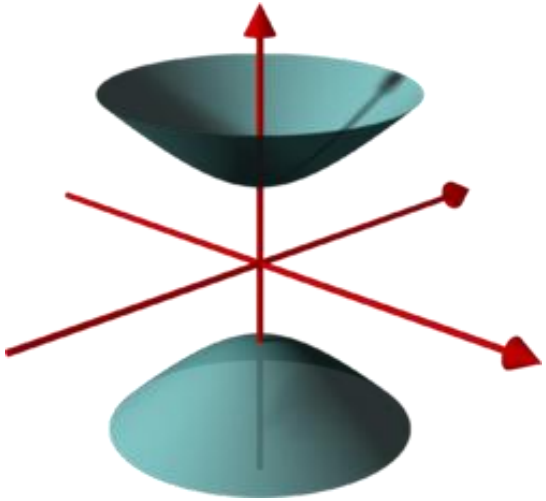
A line embedded in the usual Euclidean Space ($\mathcal{H} \subset \mathbb{R}^2$)

When adding more dimensions, we can see that the equation becomes:

$$\mathcal{H}^n: x_1^2 + x_2^2 + \cdots + x_n^2 - x_{n+1}^2 = -1$$

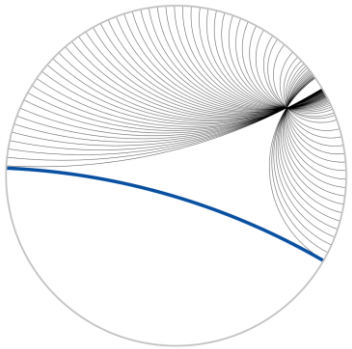
Thus

$$\mathcal{H}^n \subset \mathbb{R}^{n+1}$$



Facts about Hyperbolic Geometry

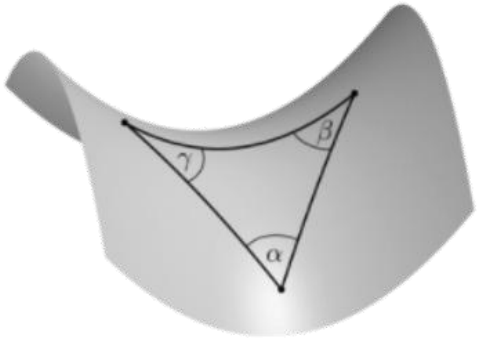
- For a line l and a point P, there are infinitely many lines through P parallel to l



- The sum of angles of a triangle is less than 180°

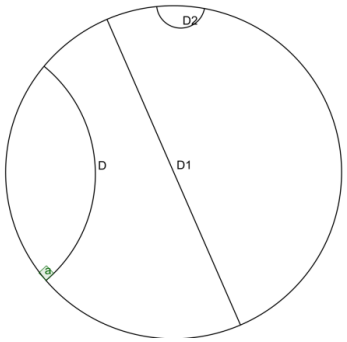
$\alpha+\beta+\gamma < 180°$

- In hyperbolic geometry, the circumference of a circle of radius r is greater than $2\pi r$

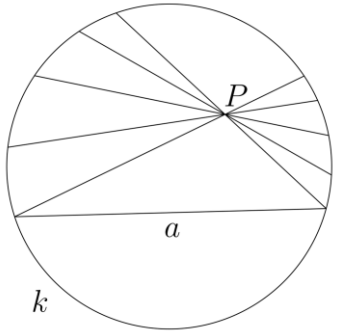


Four Models for Hyperbolic Geometry

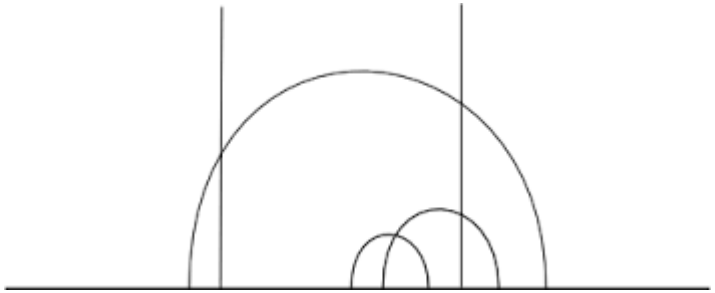
Poincaré ball model



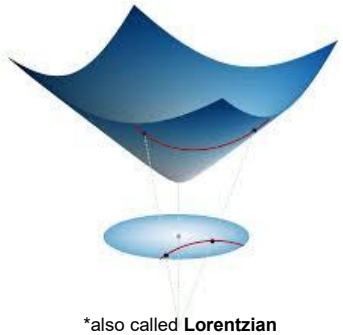
The Beltrami–Klein model



The Poincaré half-plane model



The hyperboloid* model



*also called **Lorentzian**

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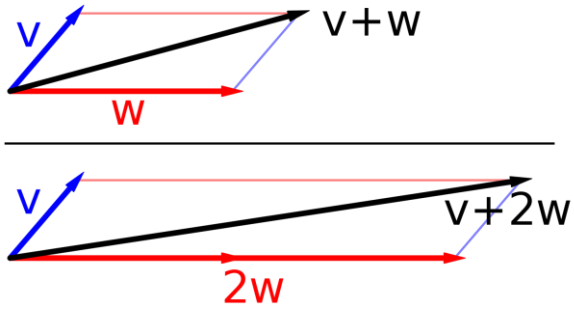
Four Models for Hyperbolic Geometry

	Pros	Cons
Poincaré ball model	Conformal	Numerically unstable
The Beltrami–Klein model	Lines are straight	Angles are not preserved
The Poincaré half-plane model	Conformal	Hard to visualize
The hyperboloid model	Numerically stable	Hard to interpret

Why not just linear algebra?

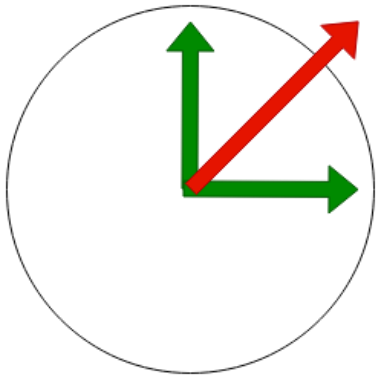
A vector space has two important operations:

- Vector addition v + w
- Scalar multiplication a * v



This does not work in hyperbolic space

We need to define something similar,
but weaker



Gyrovector space

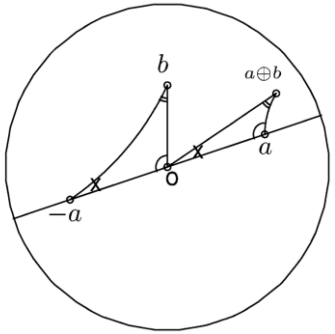


Abraham A. Ungar

Ungar replaced vector addition and scalar multiplication by two new operations:

- Möbius addition:

$$x \oplus_c y = \frac{(1 + 2c\langle x, y\rangle + c||y||^2)x + (1 - c||x||^2)y}{1 + 2c\langle x, y\rangle + c^2||x||^2||y||^2}$$



- Möbius scalar multiplication:

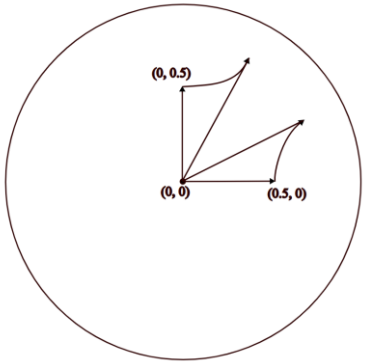
$$r \otimes_c x = \frac{1}{\sqrt{c}} \tanh\left(r \tanh^{-1}(\sqrt{c}||x||)\right) \frac{x}{||x||}$$

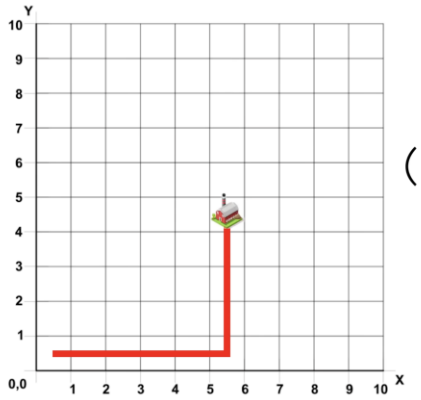
A nonlinear scaling with some nice vector space-like properties

Gyrovector space

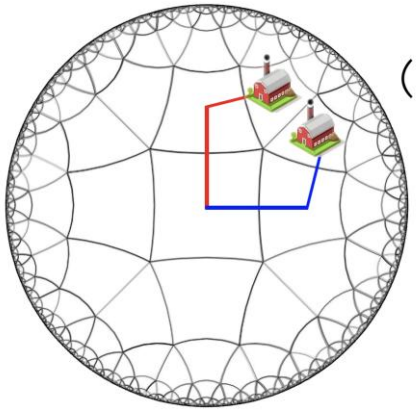
So why is it weaker? We lose commutativity, so: $x \oplus_c y \neq y \oplus_c x$

Example:





(5,4)



(1,1)?

It turns out that we can still do a lot without this commutativity

The Poincaré Ball Model

The Poincaré model can be derived using a stereoscopic projection of the hyperboloid model onto the unit circle of the $z = 0$ plane. We'll stick with the 2D case but the same ideas can be extended into higher dimensions. Figure 11 shows a visualization.

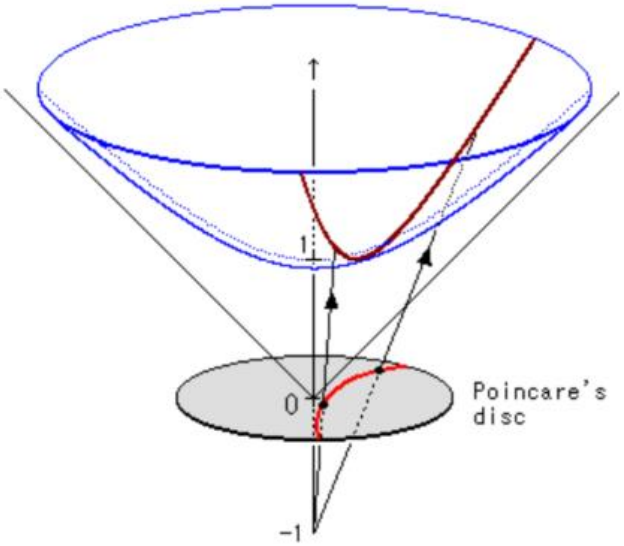


Figure 11: Stereoscopic projection to derive the Poincaré Disk Model (source: inspirehep.net).

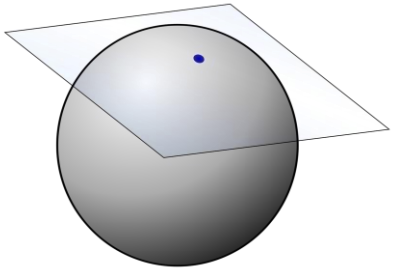
The basic idea of this stereoscopic projection (in 3D but this idea generalizes) is:

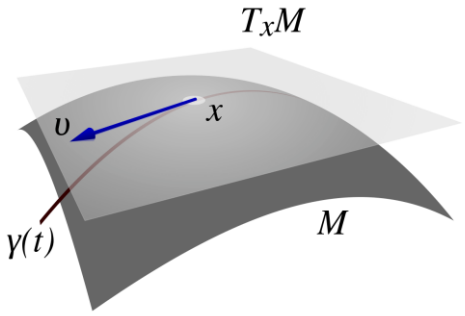
1. Start with a point P on the hyperboloid we wish to map.
2. Extend P out to a focal point $N = (0, 0, -1)$ to form a line.
3. Project that line onto the $z = 0$ plane to find our point Q in the Poincaré model.

Tangent space

The tangent space at a point of a manifold can be seen as the collection of lines that are tangent to the manifold at this point

The tangent space at a point on a sphere is a plane





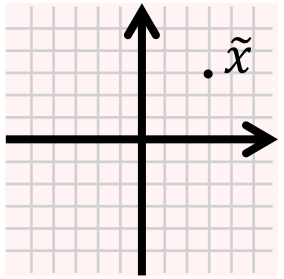
Useful for defining directions and velocities

It has Euclidean geometry

Exponential map and logarithmic map

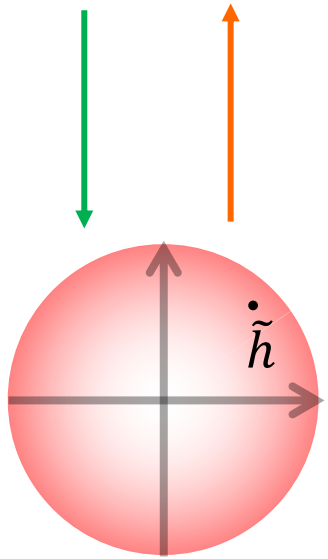
The **expmap** maps from the tangent space at a point x to the manifold itself

$$\mathrm{Exp}_x(v) = x \oplus \left(\tanh\left(\frac{\lambda_x ||v||}{2}\right) \frac{v}{||v||} \right),$$



The **logarithmic** map at a point x does the exact opposite

$$\mathrm{Log}_x(y) = \frac{2}{\lambda_x} \operatorname{artanh}\left(||-x \oplus y||\right) \frac{-x \oplus y}{||-x \oplus y||}.$$

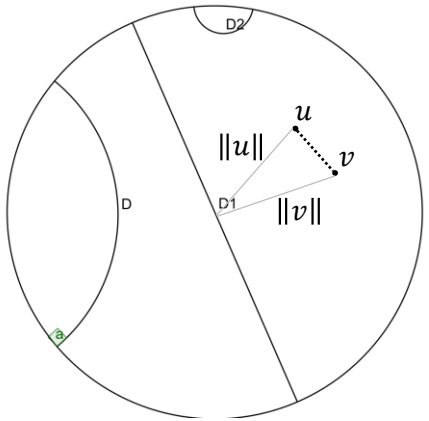


Poincaré Ball Model

- Points lie within a disc of radius 1

$$\mathbb{B}^n = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| < 1\}$$

- Straight lines (geodesics) are the diameters and the circular arcs perpendicular to the boundary



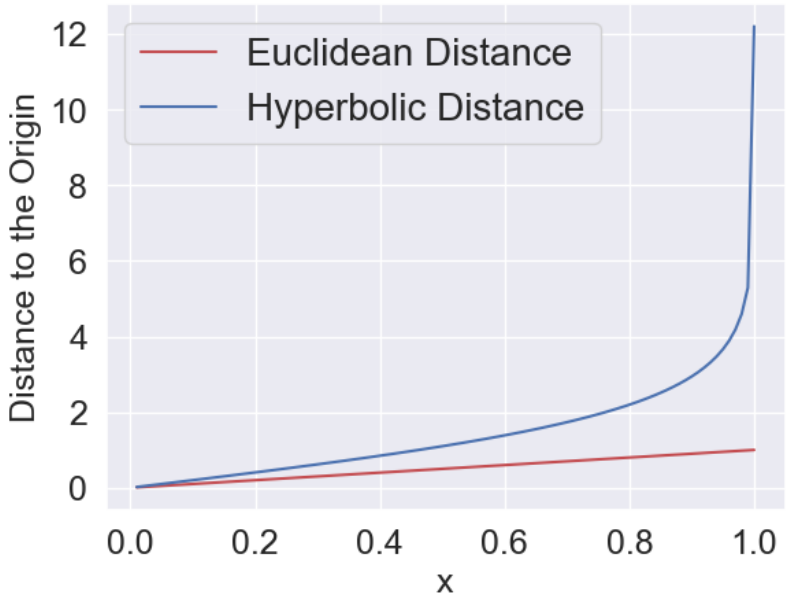
- The Poincaré distance between u and v depends *also* on the hyperbolic radii of $\|u\|$ and $\|v\|$

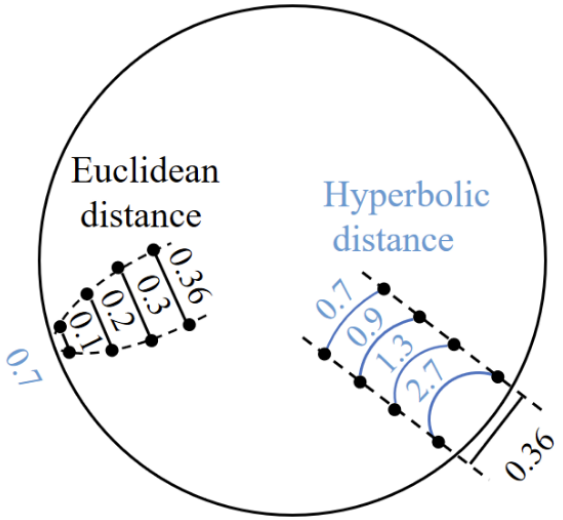
$$d_{\mathbb{B}^n}(\boldsymbol{u}, \boldsymbol{v}) = \operatorname{arcosh}\left(1 + 2\frac{\|\boldsymbol{u} - \boldsymbol{v}\|^2}{(1 - \|\boldsymbol{u}\|^2)(1 - \|\boldsymbol{v}\|^2)}\right)$$

Distance in the Poincaré Ball Model

- Sample distance plot (assuming v is in the origin)

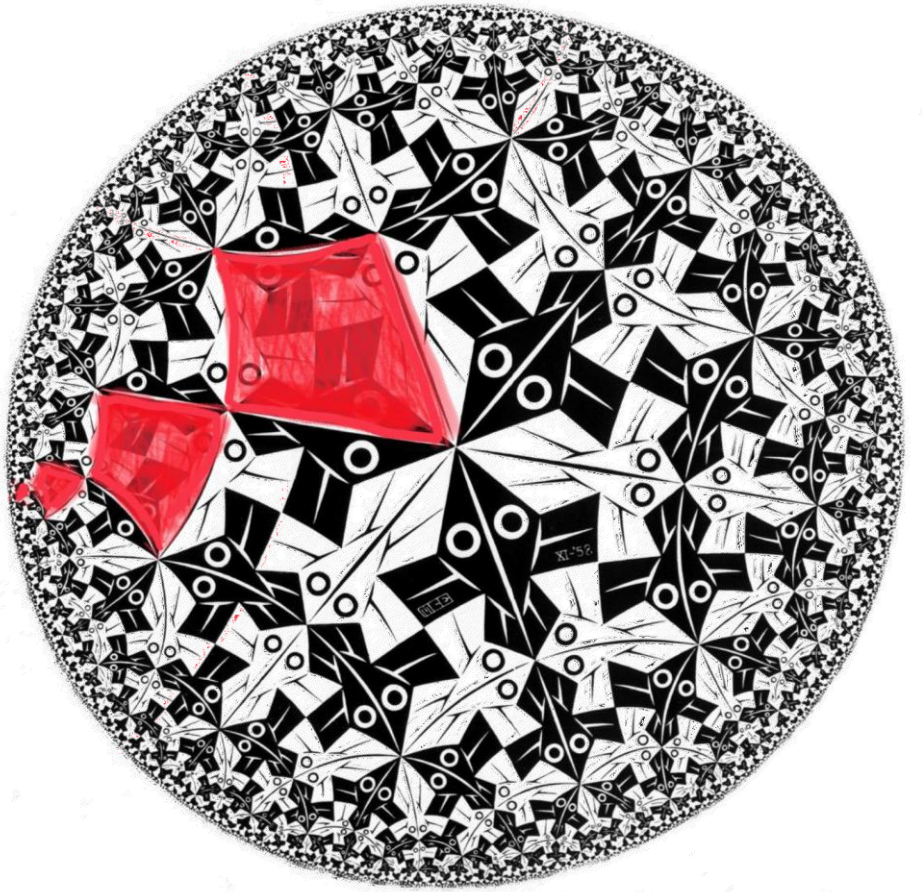
$$d_{\mathbb{B}^n}(\boldsymbol{u}, \boldsymbol{v}) = \operatorname{arcosh}\left(1 + 2\frac{\|\boldsymbol{u} - \boldsymbol{v}\|^2}{(1 - \|\boldsymbol{u}\|^2)(1 - \|\boldsymbol{v}\|^2)}\right)$$





Visual Illustration of the Poincaré Ball

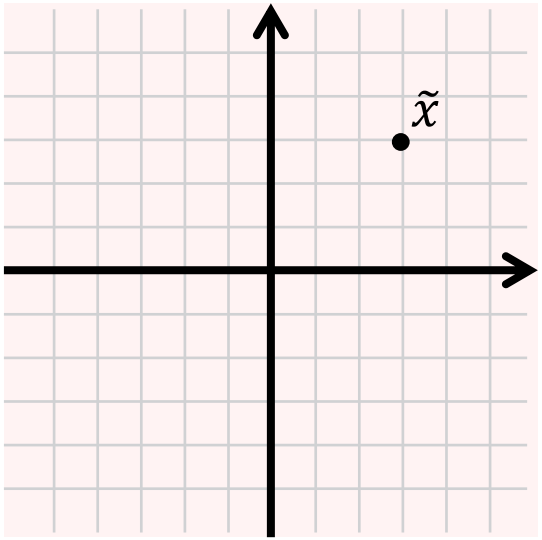
- Geodesic lines
  - the diameters and the circular arcs perpendicular to the boundary
- Regions of equal area
  - Volume grows exponentially when moving towards the ball edge



M. C. Escher (1958)

In a Nutshell

- From Euclidean to the Poincaré ball



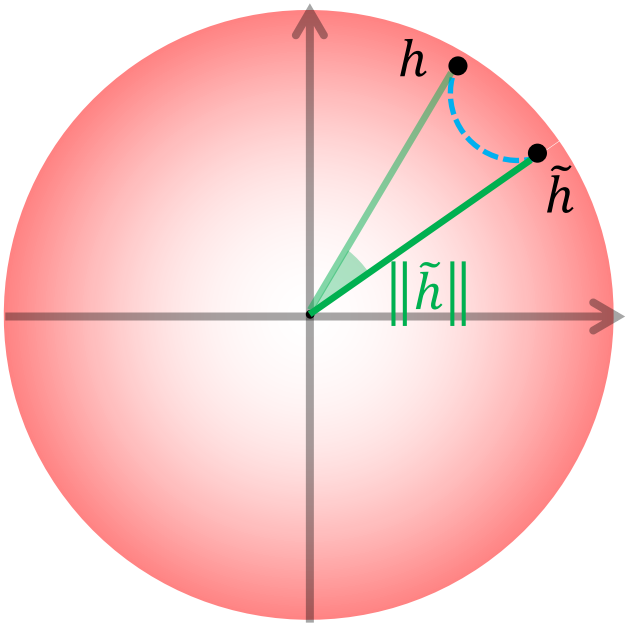
In a Nutshell

- From Euclidean to the Poincaré ball
  - One-to-one mapping:

$$\tilde{h} = Exp_0^c(\tilde{x}) = tanh(\sqrt{c}\,\|\tilde{x}\|)\frac{\tilde{x}}{\sqrt{c}\,\|\tilde{x}\|}$$ (curvature c)

  - Conformal: angles are preserved
  - Poincaré distance:

$$d_{\mathbb{D}}(h,\widetilde{h}) = \cosh^{-1}\left(1 + 2\frac{\|h-\widetilde{h}\|^2}{(1-\|h\|^2)(1-\|\widetilde{h}\|^2)}\right)$$



- Relevant to the application of the Poincaré Ball
  - Exponentially growing volume and distances allow to embed hierarchies with low distortion
  - The exponential growth of the Poincaré distance with the hyperbolic radius makes the hyperbolic radius $\|\tilde{h}\|$ a proxy for uncertainty

Brief History of Hyperbolic Studies



Nikolay Ivanovich Lobachevsky



Carl Friedrich Gauss

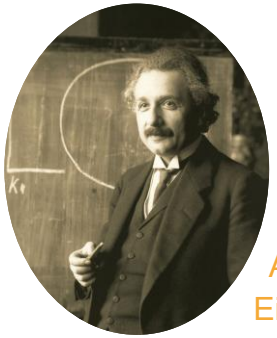


János Bolyai





Euclid



Albert Einstein



Abraham A. Ungar

HNN

- First introduced by Ganea *et al.* (2018)
- They exploit the PB model
- They formalize all the basic operations
- They propose a Hyperbolic version of the Euclidean MLR
- They define the first Computer Vision networks exploiting Hyperbolic Latent spaces



Hyperbolic Neural Networks

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Abstract

Hyperbolic spaces have recently gained momentum in the context of machine learning due to their high capacity and tree-likeliness properties. However, the representational power of hyperbolic geometry is not yet on par with Euclidean geometry, mostly because of the absence of corresponding hyperbolic neural network layers. This makes it hard to use hyperbolic embeddings in downstream tasks. Here, we bridge this gap in a principled manner by combining the formalism of Möbius gyrovector spaces with the Riemannian geometry of the Poincaré model of hyperbolic spaces. As a result, we derive hyperbolic versions of important deep learning tools: multinomial logistic regression, feed-forward and recurrent neural networks such as gated recurrent units. This allows to embed sequential data and perform classification in the hyperbolic space. Empirically, we show that, even if hyperbolic optimization tools are limited, hyperbolic sentence embeddings either outperform or are on par with their Euclidean variants on textual entailment and noisy-prefix recognition tasks.

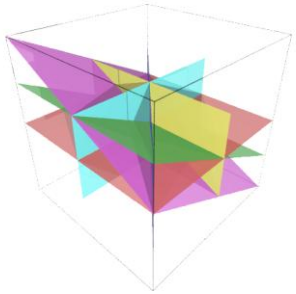
1 Introduction

It is common in machine learning to represent data as being embedded in the Euclidean space $\mathbb{R}^n$. The main reason for such a choice is simply convenience, as this space has a vectorial structure, closed-form formulas of distance and inner-product, and is the natural generalization of our intuition-friendly, visual three-dimensional space. Moreover, embedding entities in such a continuous space allows to feed them as input to neural networks, which has led to unprecedented performance on a broad range

Geometric Interpretation of MLR

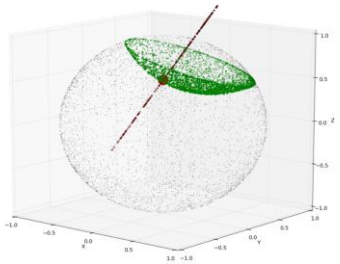
Euclidean logits in MLR: $$Ax - b = (\langle a_k, x \rangle - b_k)_k$$

Signed distance to a hyperplane: $$\frac{\langle a, x \rangle - b}{||a||}$$



Logits in MLR are proportional to scaled signed distances of the input to hyperplanes

Ganea et al. 2018



We can do the same in other geometries

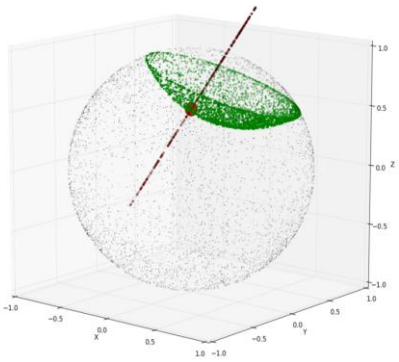
But we need a way to compute these distances

Hyperbolic MLR formulation

For each of *n* classes we have parameters:

$$p_k \in \mathbb{D}_c^n,\ a_k \in T_{p_k}\mathbb{D}_c^n \setminus \{\mathbf{0}\}:$$

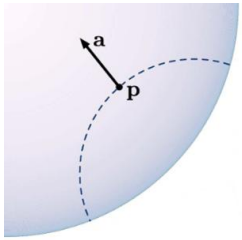
Ganea et al. 2018



With that we can compute the output probabilities as:

$$p(y=k|x) \propto \exp\left(\underbrace{\frac{\lambda_{p_k}^c \|a_k\|}{\sqrt{c}} \sinh^{-1}\left(\frac{2\sqrt{c}\langle -p_k \oplus_c x, a_k\rangle}{(1-c\|-p_k\oplus_c x\|^2)\|a_k\|}\right)}\right)$$

Scaled signed distances to hyperplanes

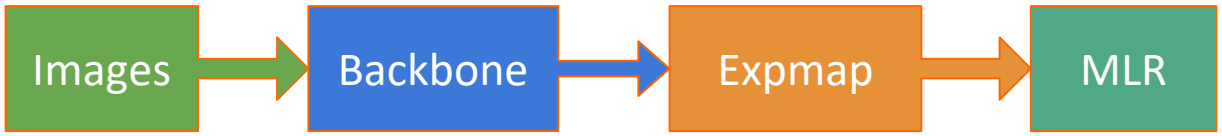


Application of MLR

Obtain hyperbolic image embeddings (upon Euclidean backbone + expmap)

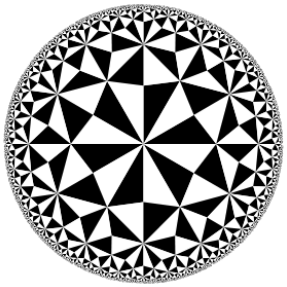
Then apply hyperbolic MLR to these hyperbolic embeddings

An example of this approach can be found in Khrulkov et al. (2020)



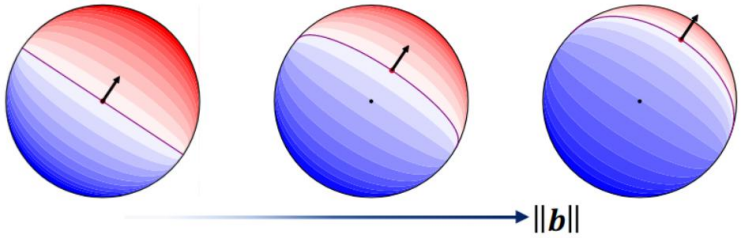
What about linear layers?

Usually defined as: $y = Ax + b$



But we do not have a notion of matrices in Hyperbolic spaces

Shimizu et al. 2020



The HNN solution:

$$y = \exp_0^c(A \log_0^c(x)) \oplus_c b$$

Activation functions

HNN applies activation functions inside the tangent space at the origin

This is not a problem as these do not explicitly use the geometry of the space

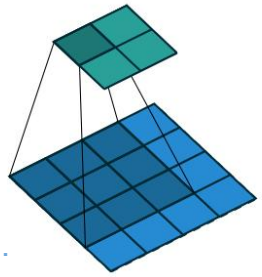
For activation function $\boldsymbol{f}$, the hyperbolic version is defined as:

$$f^{\otimes_c}(x) := \exp_{\mathbf{0}}^c(f(\log_{\mathbf{0}}^c(x)))$$

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HNN++ convolutions

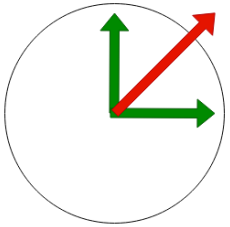
We can use linear layers to define convolutional layers, but…



Dumoulin et al. 2018

How should we concatenate the vectors in a receptive field?

Normal concatenation cannot work



HNN++ solution: β-concatenation, which maintains the mean of the norm of the inputs

β-concatenation

This type of concatenation with inputs $\{\boldsymbol{x}_i \in \mathbb{B}_c^{n_i}\}_{i=1}^N$ consists of 3 steps:

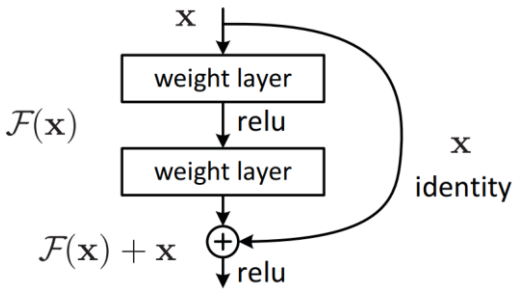
Map to tangent space at 0: $x_i \to v_i = \log_0^c(x_i) \in \mathcal{T}_0\mathbb{B}_c^{n_i}$

Scale and concatenate tangent vectors: $v := (\beta_n\beta_{n_1}^{-1}v_1^T, \ldots, \beta_n\beta_{n_N}^{-1}v_N^T)^T$

Map concatenated vectors to the Poincaré ball: $y = \exp_0^c(v)$

Poincaré ResNet

He et al. 2015

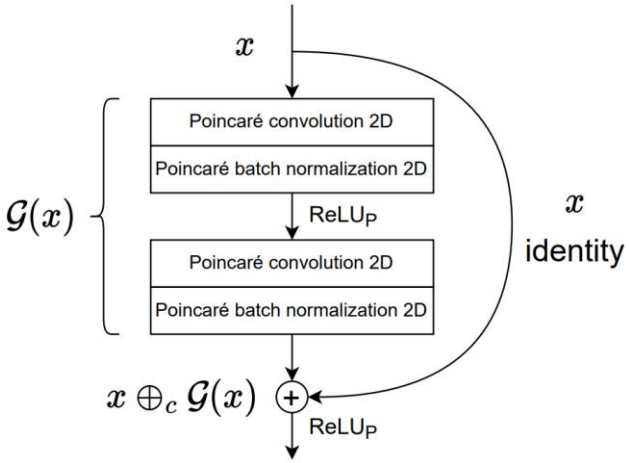


Van Spengler et al. (2023) build a fully hyperbolic ResNet

They replace the linear layers and convolutions with the HNN++ versions

They replace the batch-norm with the version by Lou et al. (2020)

They replace the skip connections by Möbius addition



HypLL

New Python hyperbolic learning library can be installed via pip

pip install hypll

Or by installing from our GitHub repository:

https://github.com/maxvanspengler/hyperbolic_learning_library

Overview

- What is hyperbolic geometry?
- Overview of the field
- From Euclid to Hyperbolic Deep Learning
  - Riemmanian optimizers
  - Hyperbolic radius
  - Entailment cones
- Leading Interpretations of the Hyperbolic Radius
  - Hyperbolic Uncertainty for Anomaly Detection
  - Hyperbolic Uncertainty for (Self-Paced) Self-Supervised Learning
- Open Research Perspectives on the Hyperbolic Radius
- Closing remarks

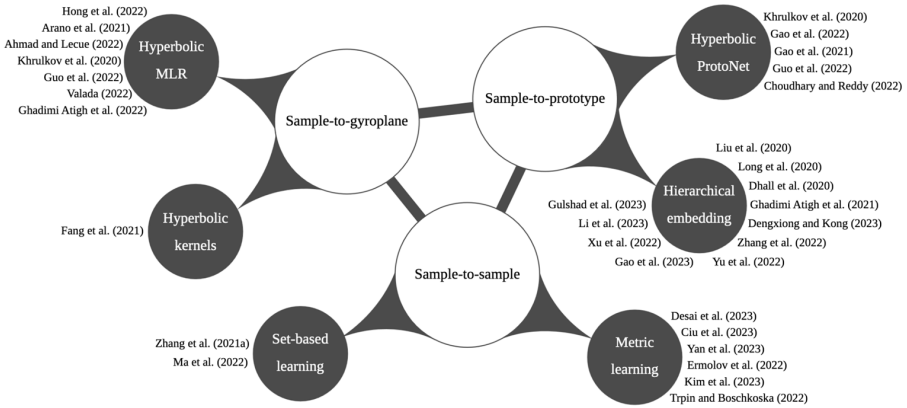
Overview of the field



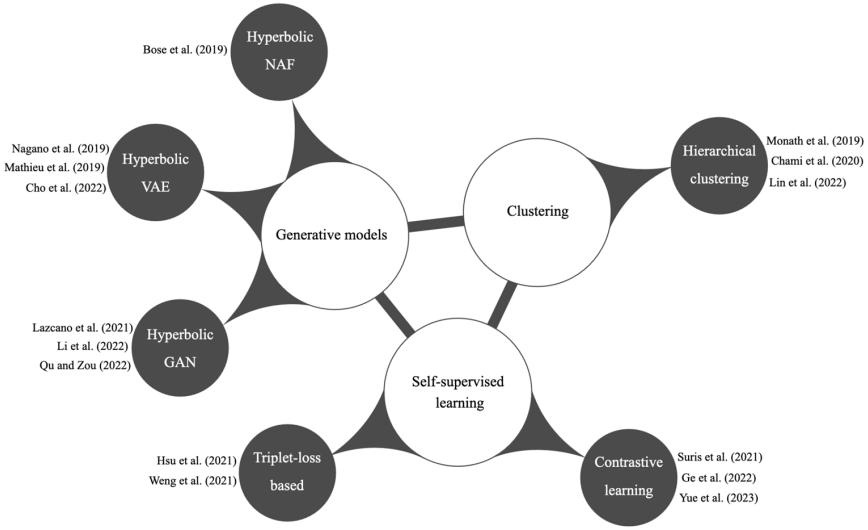
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How far are we with hyperbolic learning in vision?

Supervised learning

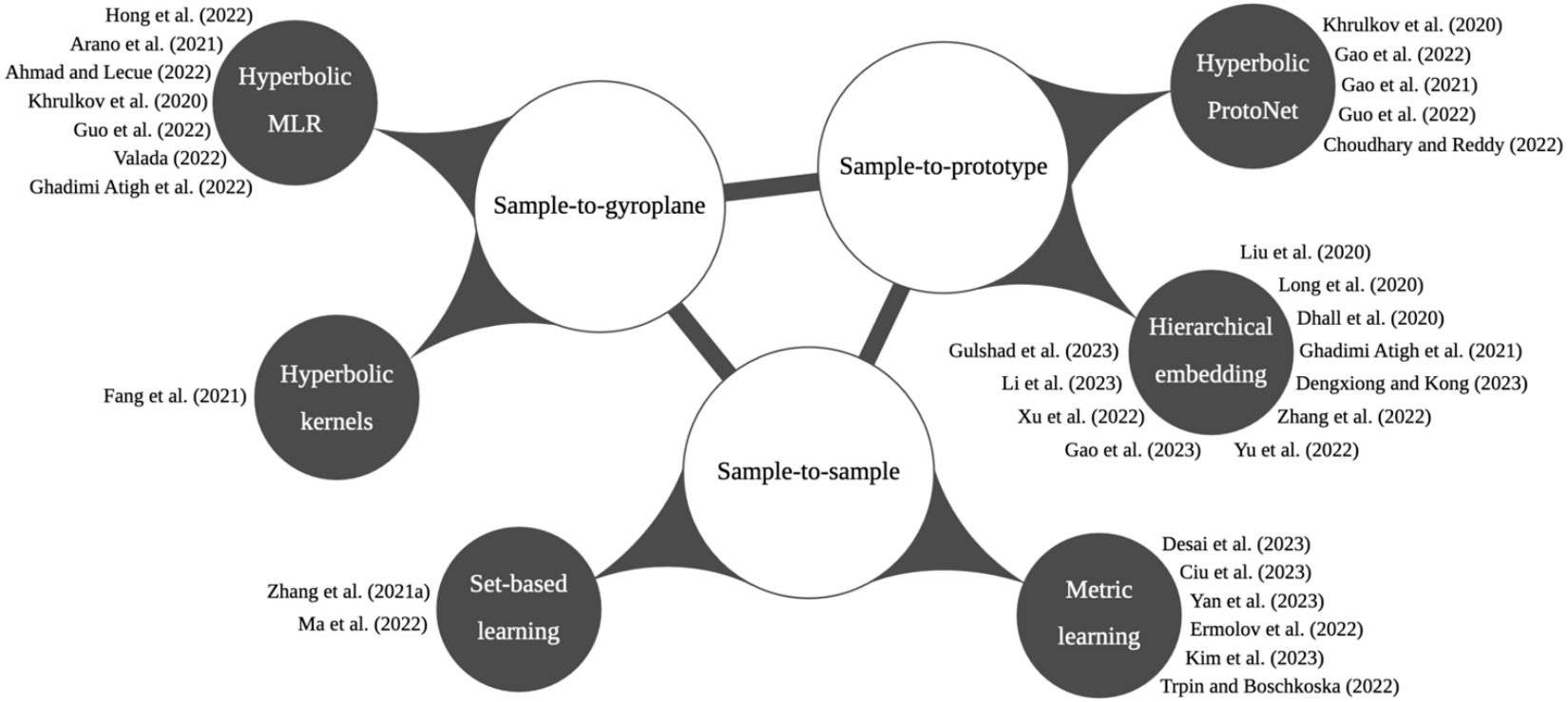


Unsupervised learning



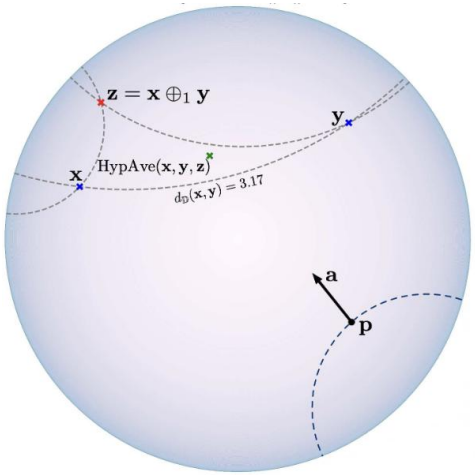
Mettes et al. Hyperbolic Deep Learning in Computer Vision: A Survey. *arXiv:2305.06611*. 2023

Supervised learning in hyperbolic space

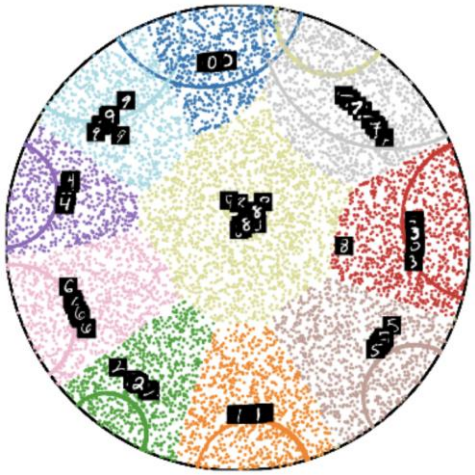


Sample-to-gyroplane learning

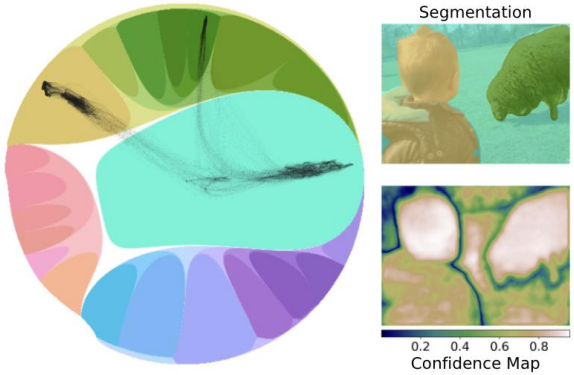
Khrulkov et al. (2020)



Guo et al. (2022)



Ghadimi Atigh et al. (2022)

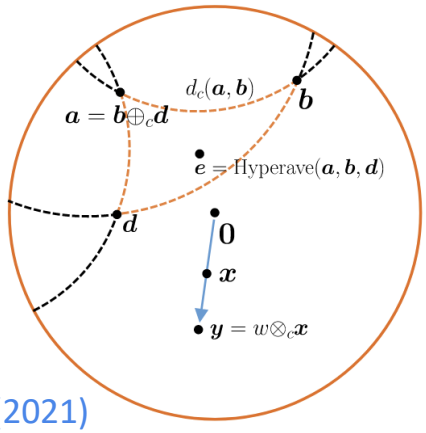


- Learn normal representations with hyperbolic logistic regression on top
- *Improves hierarchical classification, robustness, structured prediction, and more*

Sample-to-prototype learning

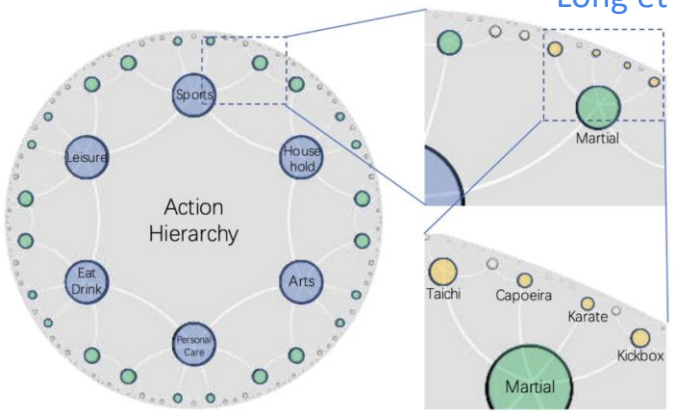
Perspective 1: prototypes from sample mean

Mostly used in few-shot learning, outperforming Euclidean conventions



Gao et al. (2021)

Long et al. (2020)

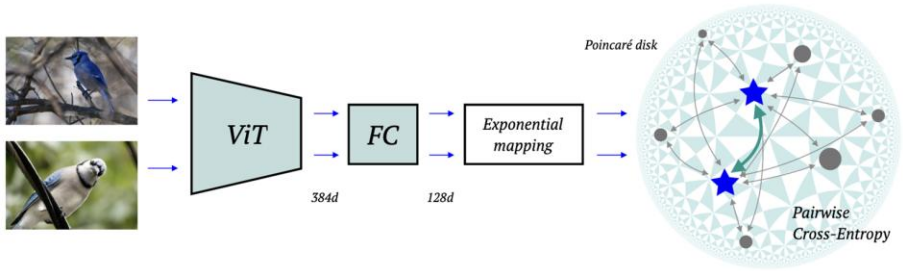


Perspective 2: prototypes from hierarchical embeddings

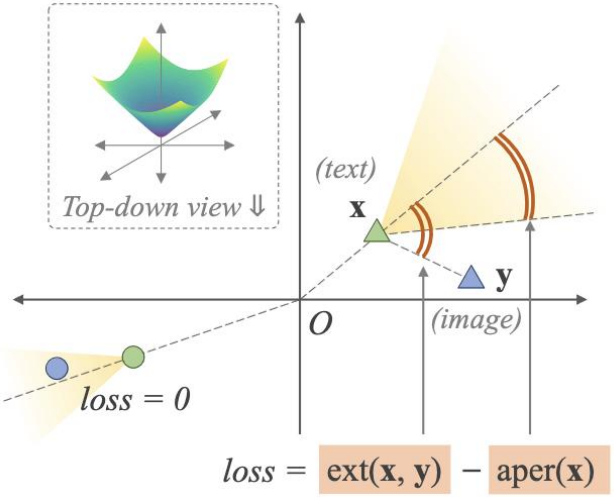
Ideal for learning with prior hierarchical knowledge and zero-shot generalization

Sample-to-sample learning

Hyperbolic Vision Transformers
Ermolov et al. (2022)

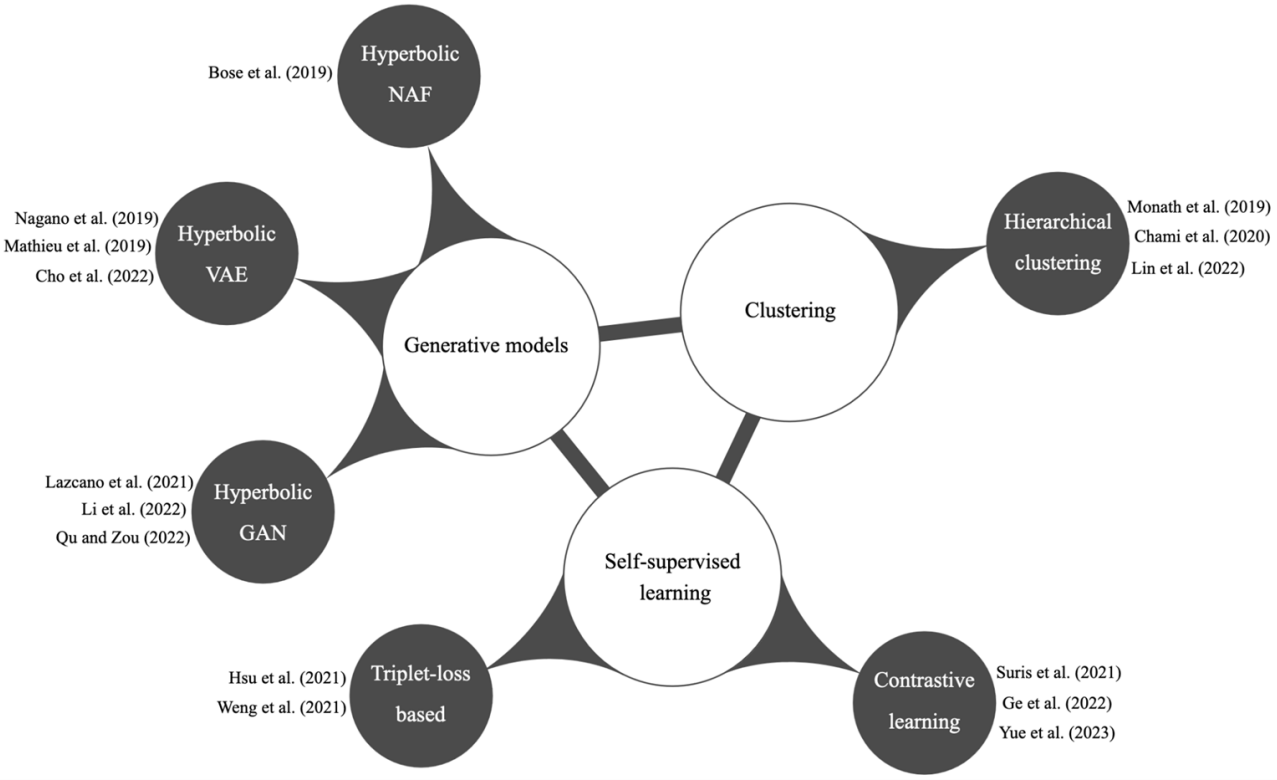


Hyperbolic CLIP
Desai et al. (2023)



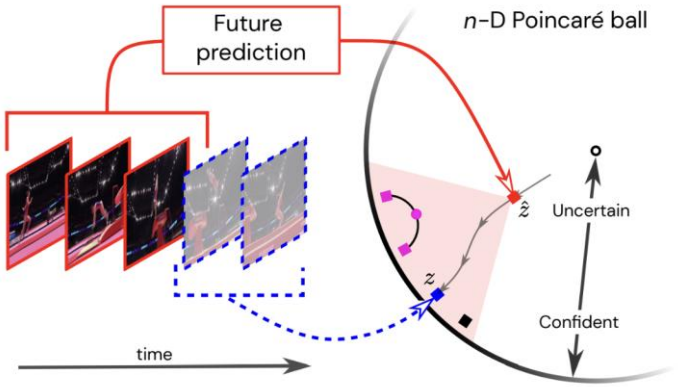
- Pull and push sample pairs akin to contrastive and metric learning in Euclidean space
- *Great for fine-grained and multi-modal tasks, even at scale*

Unsupervised learning in hyperbolic space

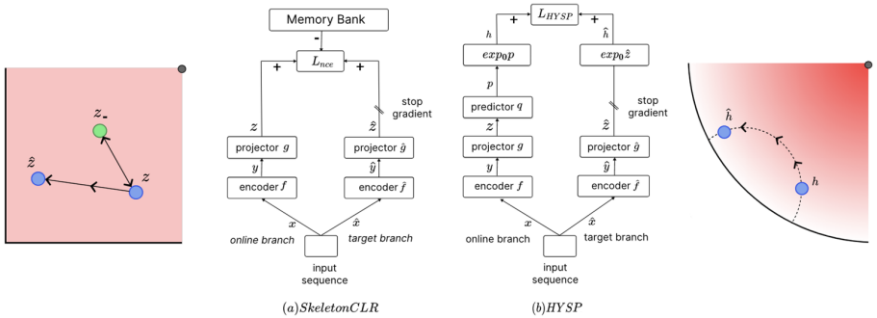


Self-supervised learning

Surís et al. (2021)

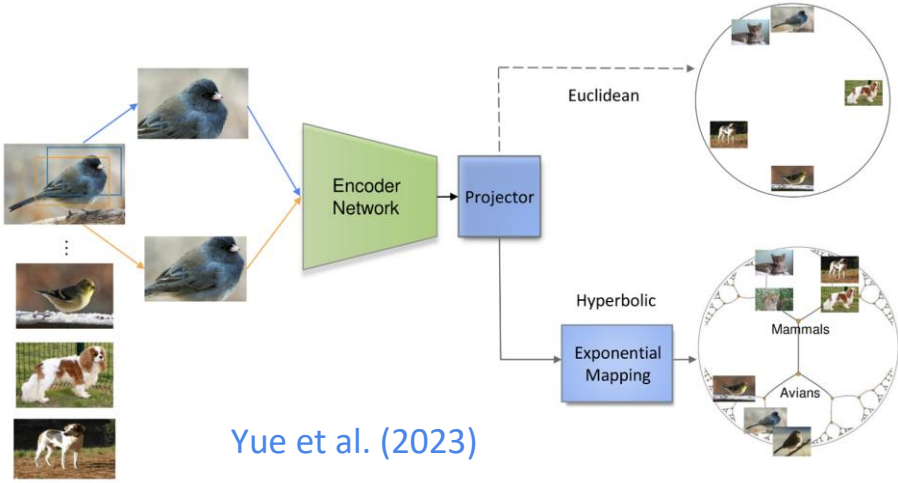


Franco et al. (2023)



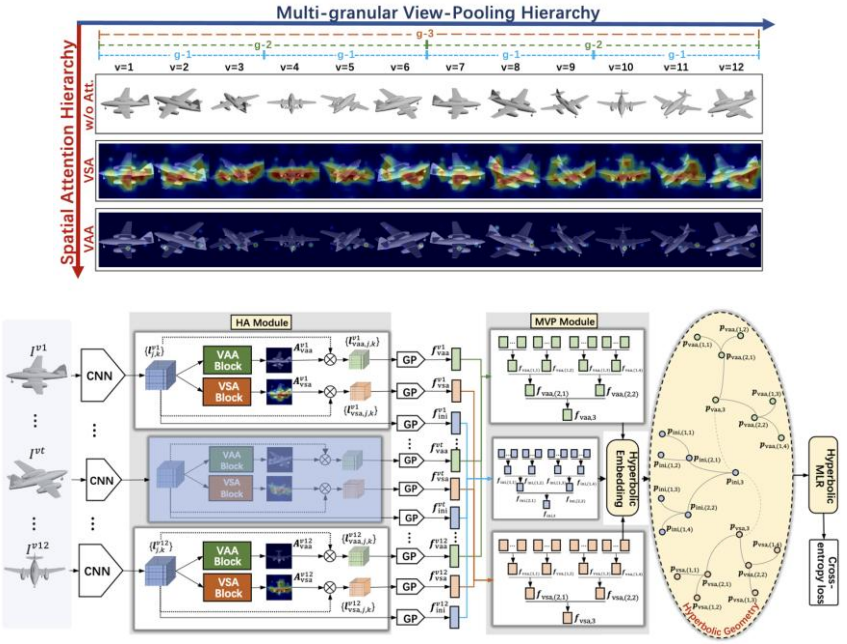
- Self-supervised objective transfer naturally to hyperbolic space
- *Hierarchical representations without labels and inherent uncertainty quantification*

Hyperbolic 3D vision



Yue et al. (2023)

Chen et al. (2020)



- Scene-object, point clouds, and LiDAR also deal with hierarchies and uncertainty
- *Hyperbolic embeddings help with supervised and self-supervised learning on 3D data*

Pros and cons

Most effective for:

- hierarchical learning
- end-to-end uncertainty estimation
- few-sample learning
- robust learning
- low-dimensional learning

Open challenges include:

- fully-hyperbolic learning
- computational hurdles
- open-source development
- large-scale learning

Overview

- What is hyperbolic geometry?
- Overview of the field
- From Euclid to Hyperbolic Deep Learning
  - Riemmanian optimizers
  - Hyperbolic radius
  - Entailment cones
- Leading Interpretations of the Hyperbolic Radius
  - Hyperbolic Uncertainty for Anomaly Detection
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- Open Research Perspectives on the Hyperbolic Radius
- Closing remarks

From Euclid to Hyperbolic Deep Learning

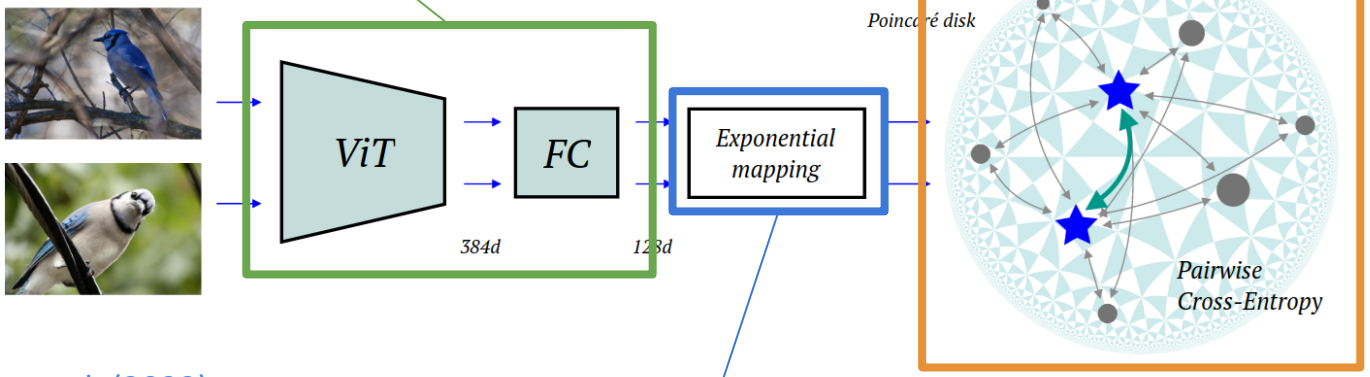


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Hyperbolic image embeddings

1. Compute image embeddings with a backbone

3. Perform downstream task



Ermolov et al. (2022)

2. Map the embeddings to the Poincaré ball

Riemmanian SGD

- Bécigneul et al. also introduce Riemannian versions of several optimizers, based on the Riemannian SGD by Bonnabel (2013)

- This connects the tools from HNN with the Riemannian optimizers and it yields hyperbolic optimizers

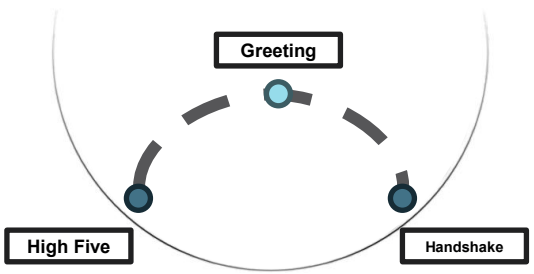
- They are the optimizers to be used when dealing with HNN!

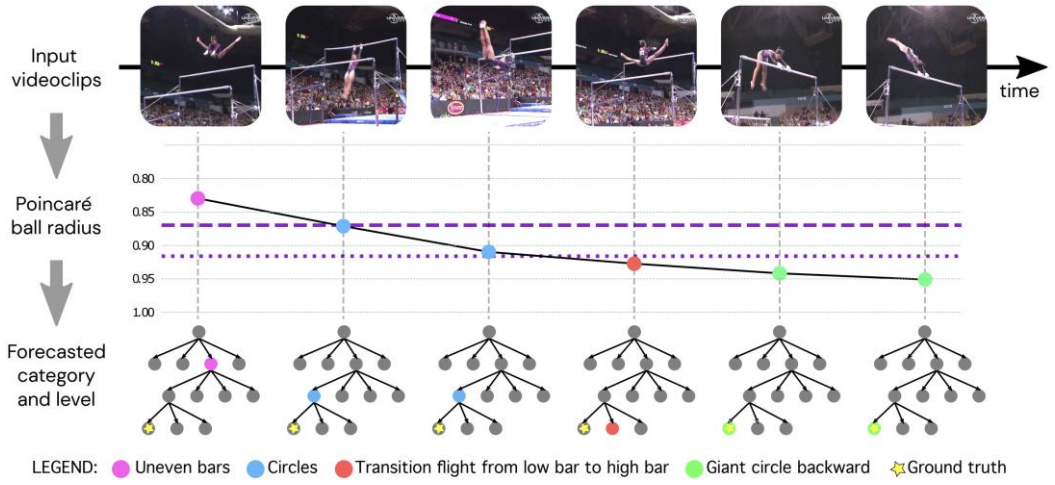
The hyperbolic radius

- Surís *et al*. defined the first NN using Hyperbolic latent spaces for action prediction
- Akin to the prototypes learning framework, they enforce similar actions to be embedded in close regions of the space via metric learning

$$\mathcal{L} = -\sum_i \left[\log \frac{\exp(-d^2_{\mathbb{D}}(\hat{z}_i, z_i))}{\sum_j \exp(-d^2_{\mathbb{D}}(\hat{z}_i, z_j))} \right]$$



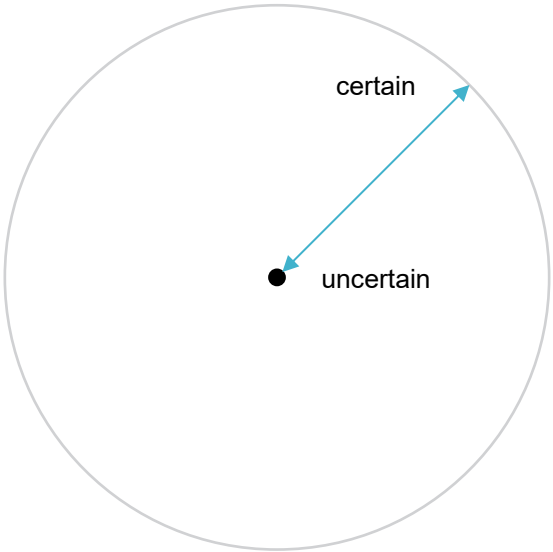




Hyperbolic Radius as Model (Un)certainty

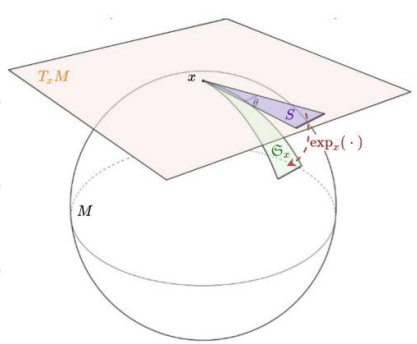
- After Surís (2021), several researchers begin to investigate this property of hyperbolic embeddings

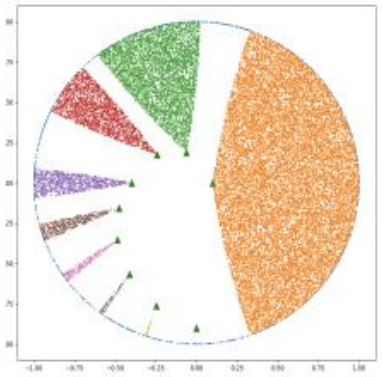
- A simple method for assessing the model uncertainty is very interesting as it can empower the model with the ability of «abstaining» in case of very «uncertain» results



Entailment Cones

- Hyperbolic spaces are the proper space to embed tree-like structures
- Taxonomies and hierarchies can be represented as tree-like graphs
- But how to embed this *entailment* relation in the network?
  - We want a geometrical sorting of the embeddings that reflects the relations between the objects in a hierarchy
- Ganea (2018) proposed to use the *Entailment Cones*



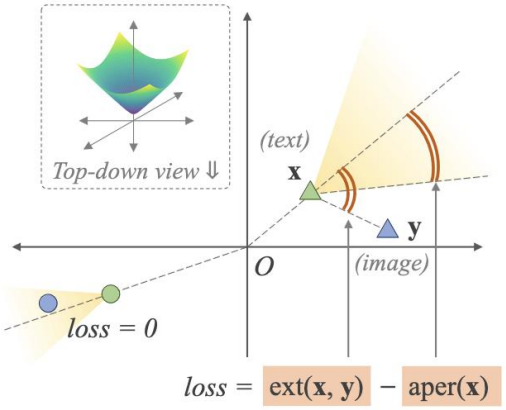


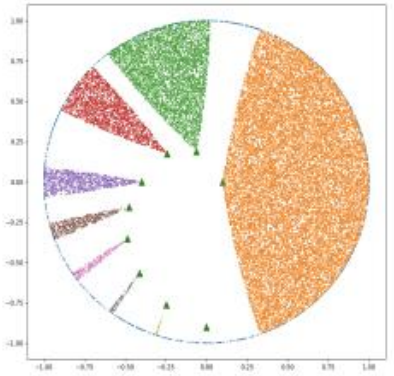
Entailment Cones

- The idea is to define conical regions in the PB that stem (have the vertex) from the embedding of the parent concept **(x)**
  - The aperture of these cones depends only on the norm of the parent embedding: $\text{aper}(\mathbf{x}) = \sin^{-1}\left(\frac{2K}{\sqrt{c}\,\|\mathbf{x}_{space}\|}\right)$

- All the child concepts **(y)** should belong to this region

- The result is achieved trough metric learning, forcing the entailed objects to rotate into the chosen region





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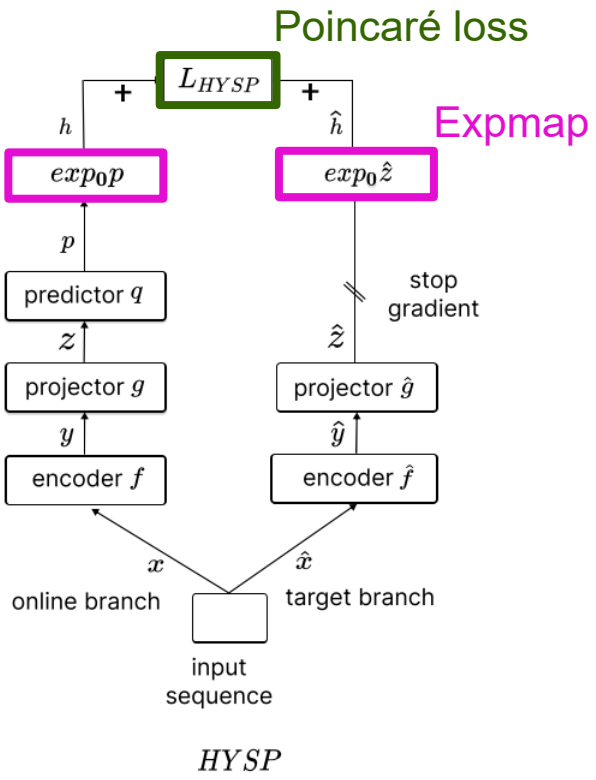
Leading Interpretations of the Hyperbolic Radius



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Hyperbolic Radius in Computer Vision

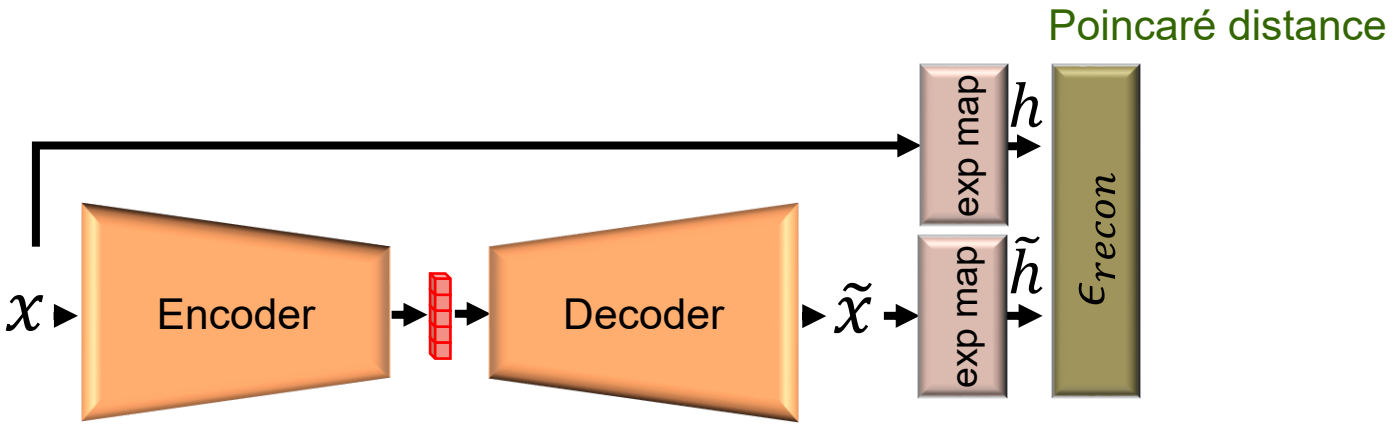
- Radius → certainty
  - Weight samples in SSL via the hyperbolic radius
    - **Franco et al. ICLR'23**



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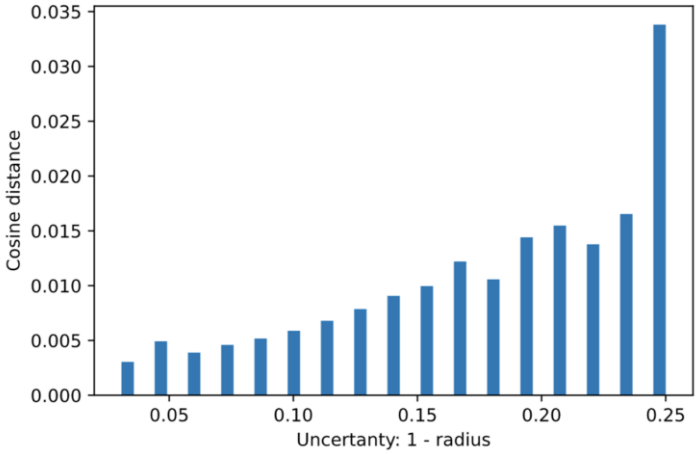
Hyperbolic Radius in Computer Vision

- Radius → certainty
  - Weight samples in SSL via the hyperbolic radius
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  - Detect anomalies and abstain if uncertain
    - **Flaborea et al. CVPR'23 Wks**

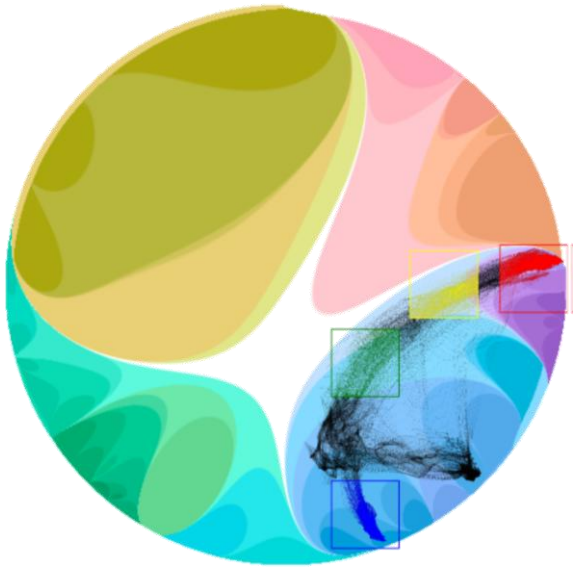


Hyperbolic Radius in Computer Vision

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  - Larger error when more uncertain



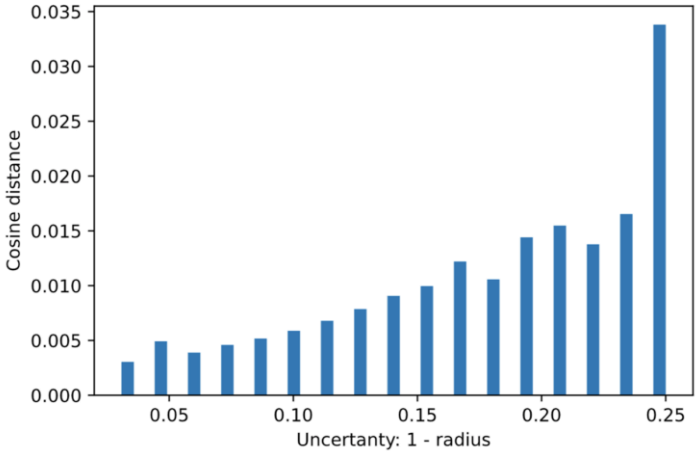
- Radius → hierarchy (parent-to-child)
  - In hierarchical classification, parent classes have lower radii
    - Ghadimi Atigh et al. CVPR'22



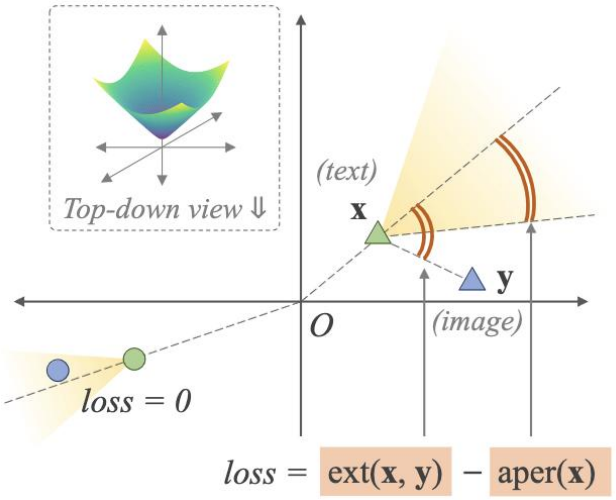


Hyperbolic Radius in Computer Vision

- Radius → certainty
  - Weight samples in SSL via the hyperbolic radius
    - **Franco et al. ICLR'23**
  - Detect anomalies and abstain if uncertain
    - **Flaborea et al. CVPR'23 Wks**
  - Larger error when more uncertain



- Radius → hierarchy (parent-to-child)
  - In hierarchical classification, parent classes have lower radii
    - Ghadimi Atigh et al. CVPR'22
  - Enforce image-text hierarchies by an entailment loss
    - Desai et al. PMLR'23



Hyperbolic Uncertainty for Anomaly Detection

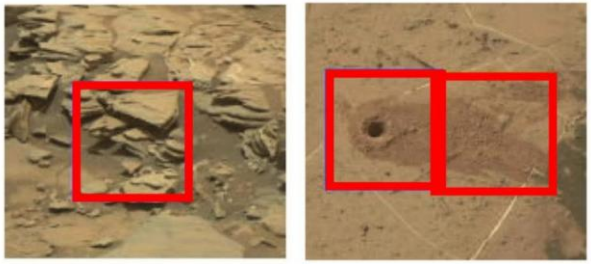
- Prenkaj, Aragona, Flaborea, Galasso, Gravina, Podo, Reda, Velardi (2023). A self-supervised algorithm to detect signs of social isolation in the elderly from daily activity sequences. In *Artificial Intelligence in Medicine*
- Flaborea, Prenkaj, Munjal, Sterpa, Aragona, Podo, Galasso (2023). Are we certain it's anomalous? In Proc. *CVPR wks*



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Anomaly Detection Applications

Rover-Based Space Exploration:
unknown textures



Bedrock
(Sol 1032)

Drill hole and tailings
(Sol 1496)

Video surveillance:
anomalous behavior, accidents, fights..

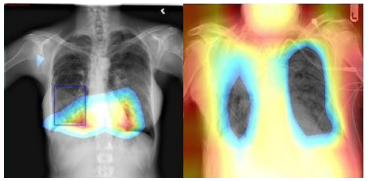


Normal Frame **Anomalous Frame**

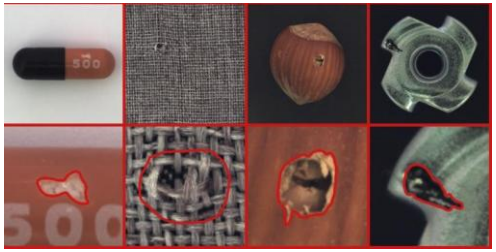
Finance:
credit card/insurance frauds, market manipulation, money laundering, etc.



Healthcare:
lesions, tumours, events in IoT/ICU monitoring, etc.



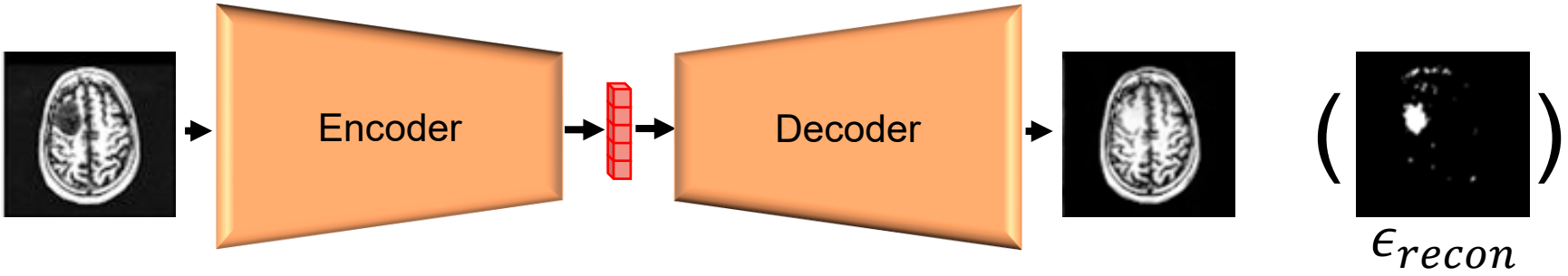
Industrial Inspection:
Defects, micro-cracks



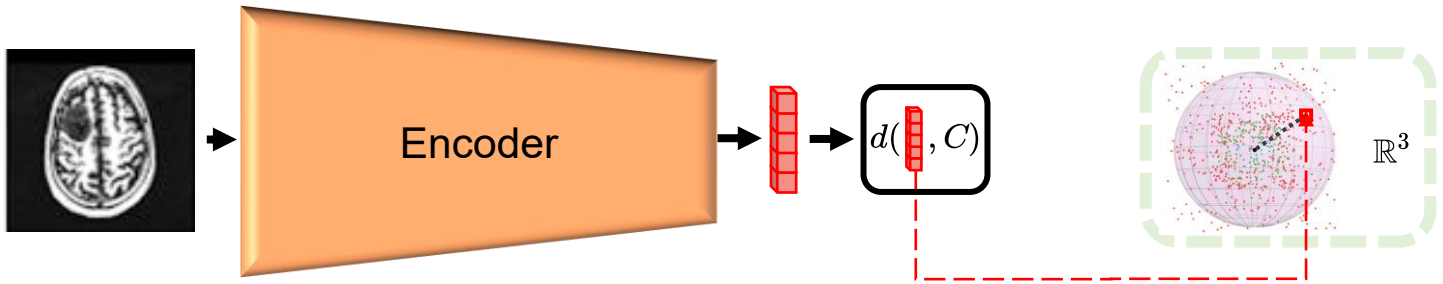
Slide credit: Guansong Pang, Longbing Cao, Charu Aggarwal

Anomaly Detection

- Learn to reconstruct normalcy, compare input Vs. reconstructed



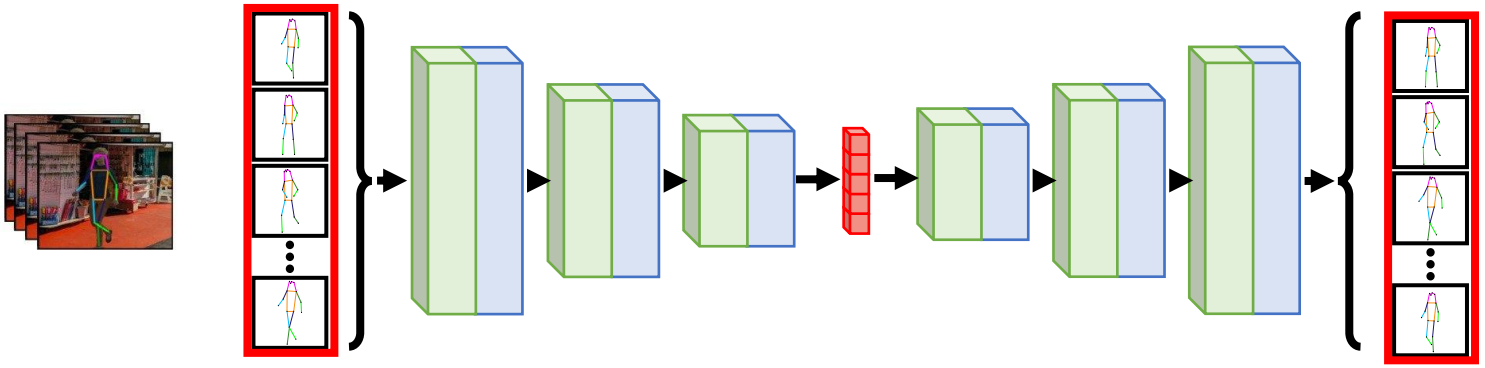
- Constrain normalcy into a hypersphere, measure dist. from center



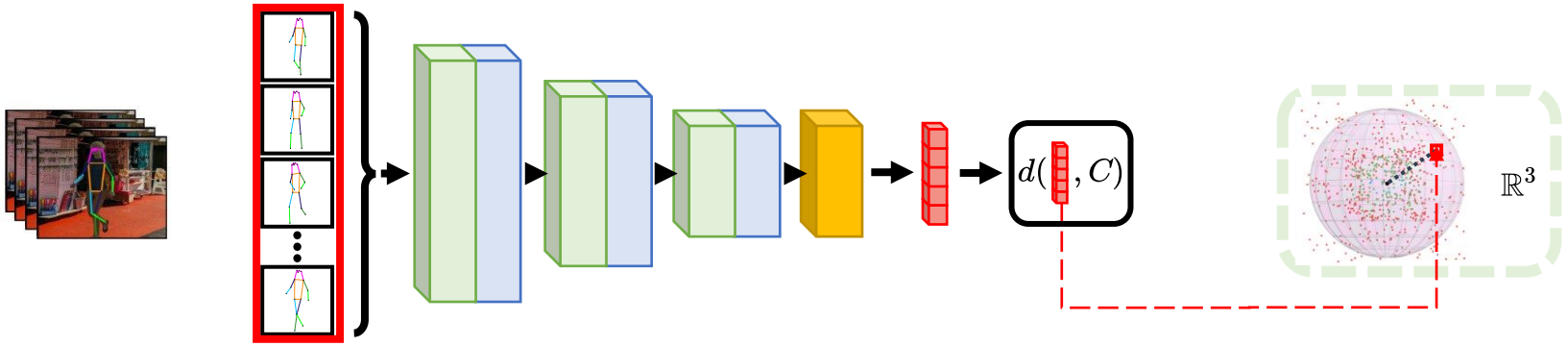
Skeletal Motion-based Anomaly Detection



- Learn to reconstruct normalcy, compare input Vs. reconstructed

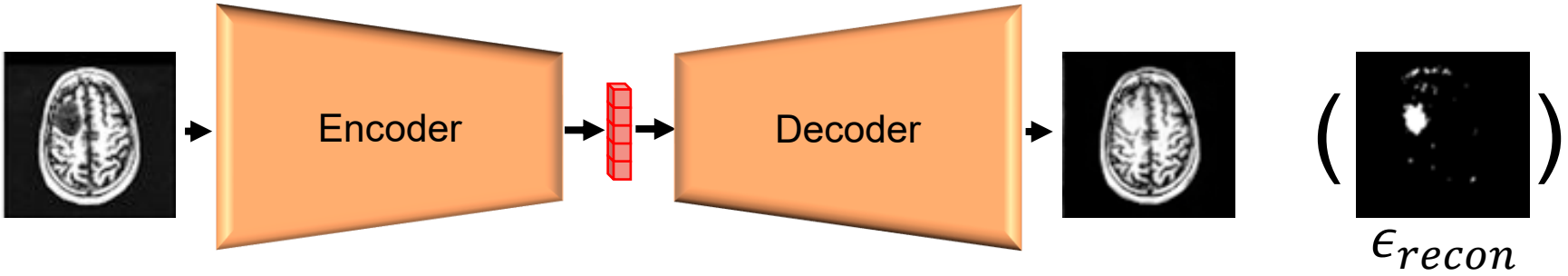


- Constrain normalcy into a hypersphere, measure dist. from center



Anomaly Detection

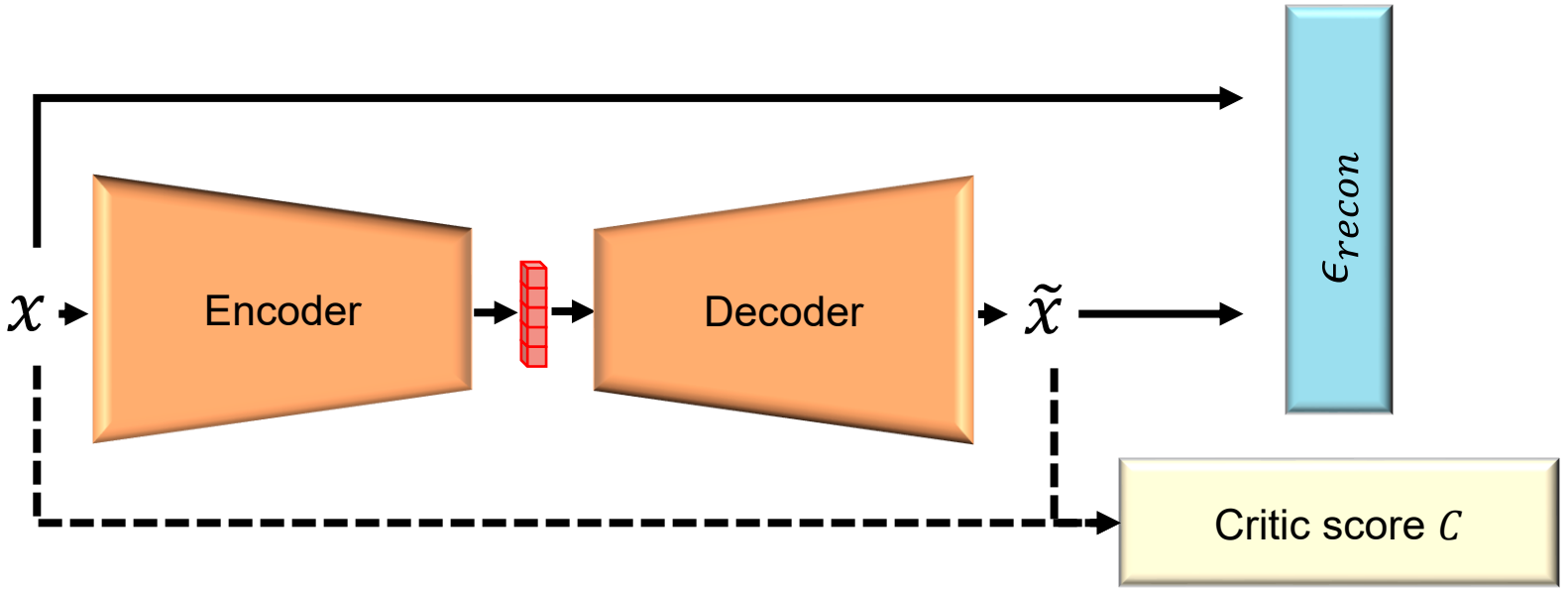
- Learn to reconstruct normalcy, compare input Vs. reconstructed



- Constrain normalcy into a hypersphere, measure dist. from center

Anomaly Detection by Reconstruction Error

- Build on the current best (TADGAN)
  - $\epsilon_{recon} = \text{Dist}(x, \tilde{x})$
  - Critic score C for adversarial training
  - anomaly $= \epsilon_{recon} * C$

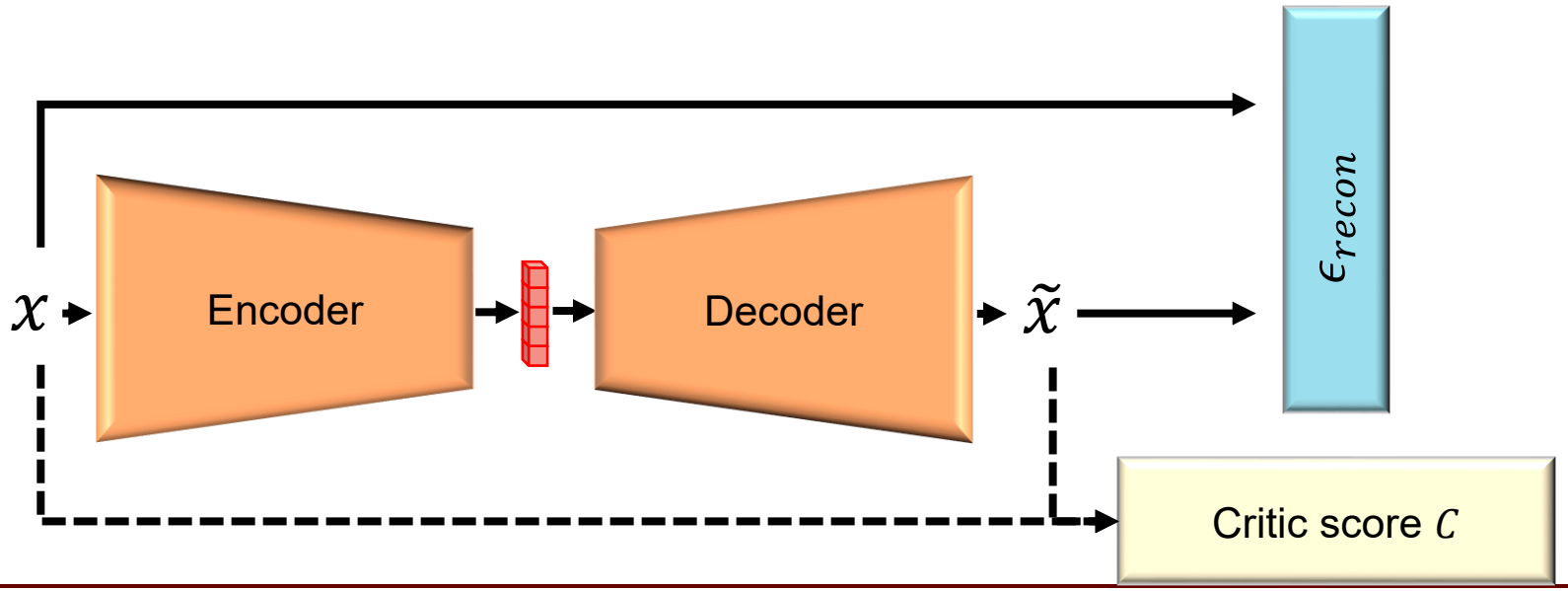


Geiger et al. (2020). "TADGAN: Time series anomaly detection using generative adversarial networks". In IEEE Int. Conf. on Big Data'20

Proposed: HypAD
Hyperbolic Uncertainty for Anomaly Detection

- Proposed: *trust* reconstruction errors if *certain* about the sample

→ estimate the uncertainty of the samples

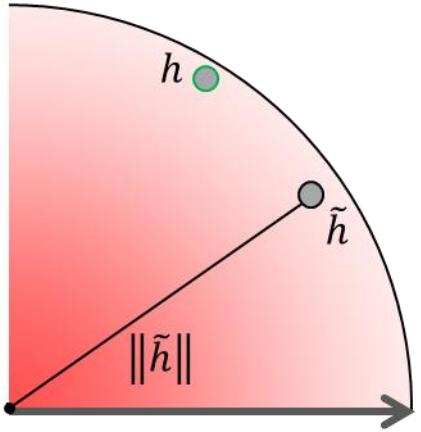


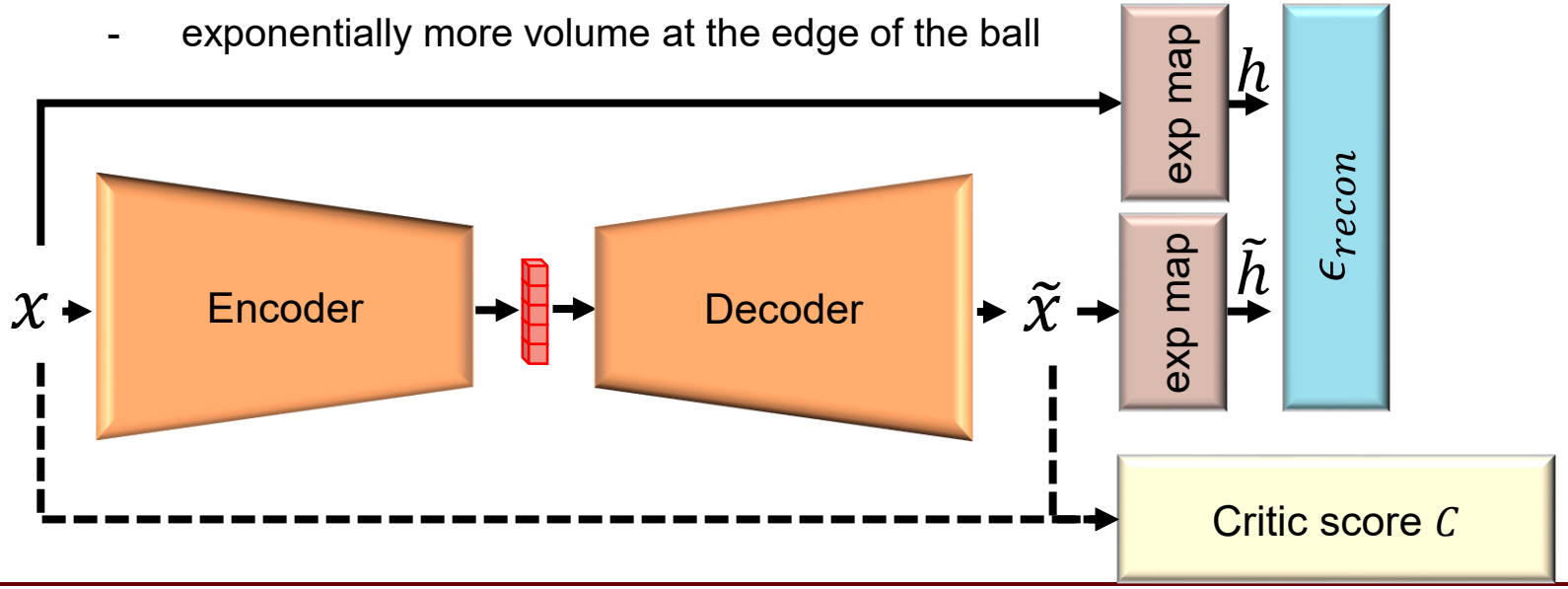
Proposed: HypAD
Hyperbolic Uncertainty for Anomaly Detection

- Minimize reconstruction errors in Hyperbolic space
  - Map x and the reconstructed $\tilde{x}$ into the Poincaré ball via exp map [Ganea et al. NeurIPS'18]

$$\tilde{h} = Exp_0^c(\tilde{x}) = tanh(\sqrt{c}\,\|\tilde{x}\|)\frac{\tilde{x}}{\sqrt{c}\,\|\tilde{x}\|}$$

  where c is the curvature
  - exponentially more volume at the edge of the ball



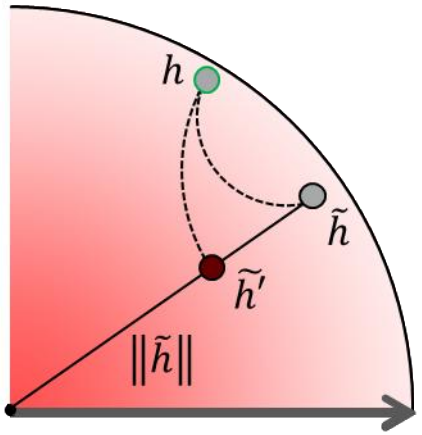


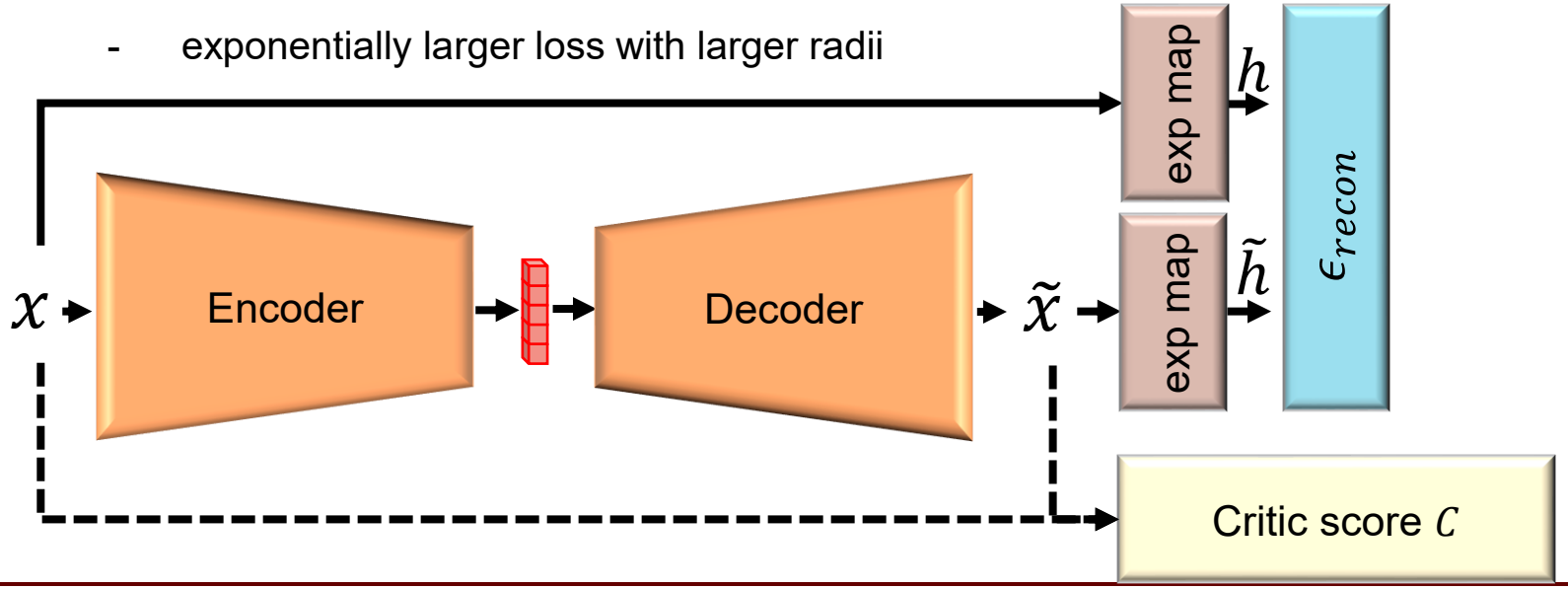
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  - Compare the embeddings via the Poincaré distance:

$$Z_{RE}(x) = \cosh^{-1}\left(1 + 2\frac{\|h - \tilde{h}\|^2}{(1 - \|h\|^2)(1 - \|\tilde{h}\|^2)}\right)$$

    - exponentially larger loss with larger radii



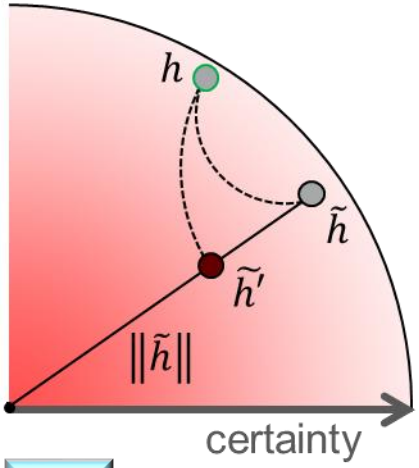


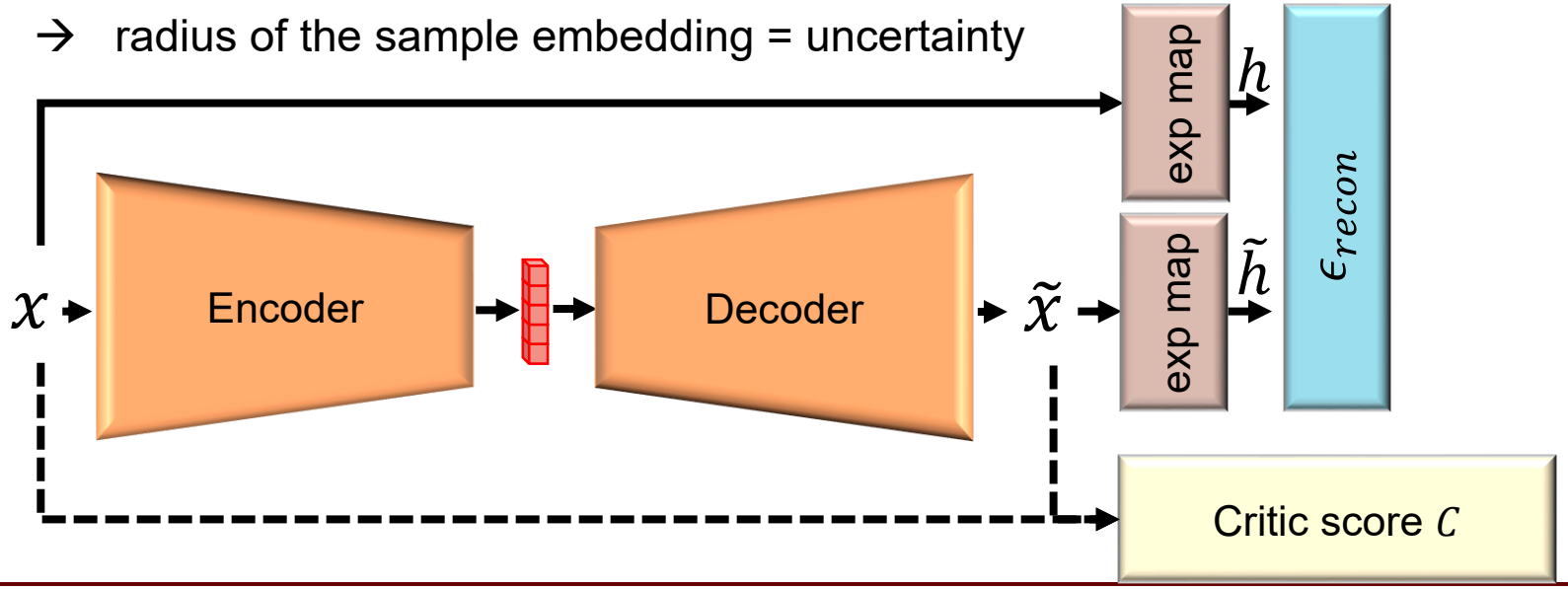
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→ radius of the sample embedding = uncertainty



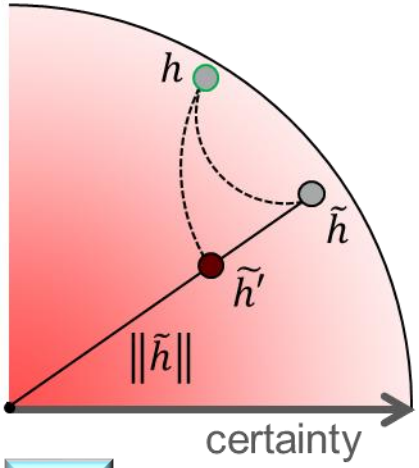


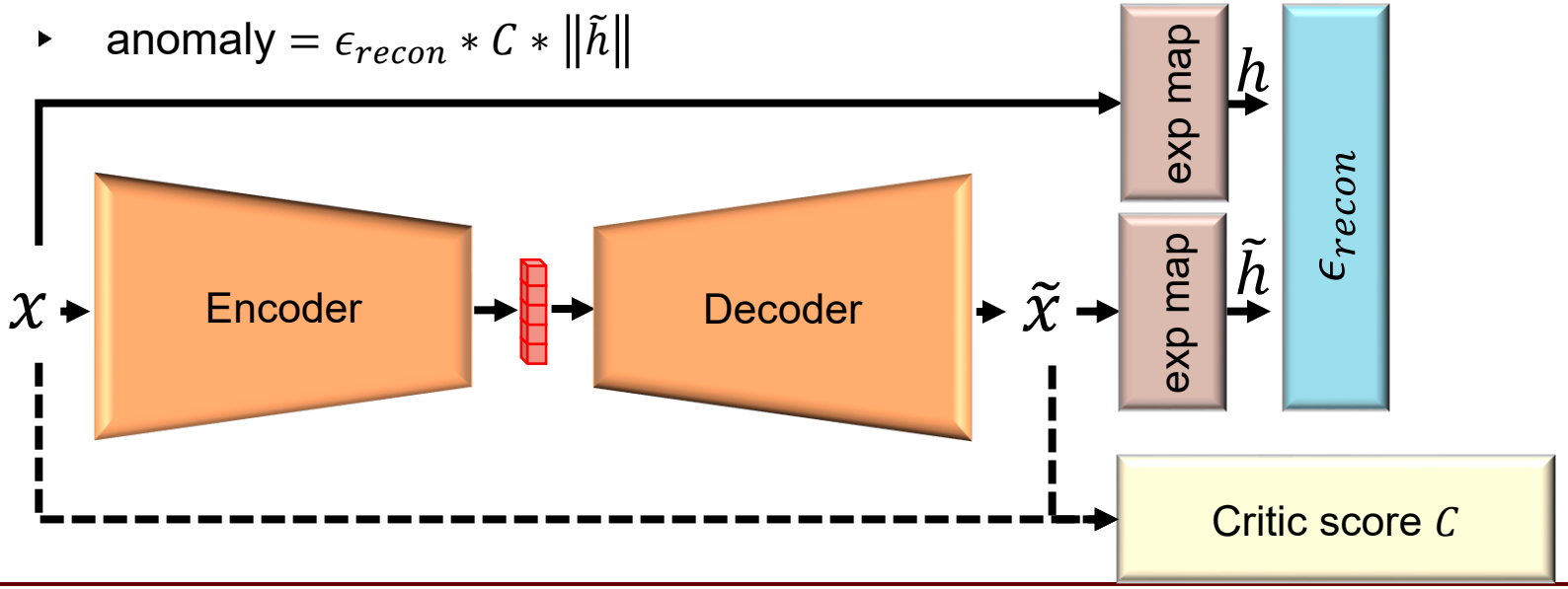
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  - anomaly $= \epsilon_{recon} * C * \|\tilde{h}\|$





Evaluation of HypAD

- Surpass SoA on established anomaly detection benchmarks:

	NASA		YAHOO				NAB					
	MSL	SMAP	A1	A2	A3	A4	Art	AdEx	AWS	Traf	Tweets	F1 ($\mu \pm \sigma$)
TadGAN [Geiger et al.(2020)]	0.623	0.680	0.668	0.820	0.631	0.497	0.667	0.667	0.610	0.455	0.605	0.629 ± 0.123
AE	0.199	0.270	0.283	0.008	0.100	0.073	0.283	0.100	0.239	0.088	0.296	0.176 ± 0.099
LstmAE	0.317	0.318	0.310	0.023	0.097	0.089	0.261	0.130	0.223	0.136	0.299	0.200 ± 0.103
ConvAE	0.300	0.292	0.301	0.000	0.103	0.073	0.289	0.129	0.254	0.082	0.301	0.212 ± 0.096
TadGAN*	0.500	0.580	**0.620**	**0.865**	**0.750**	**0.576**	0.420	0.550	**0.670**	0.480	0.590	0.600 ± 0.115
HypAD (proposed)	**0.565**	**0.643**	0.610	0.670	0.670	0.470	**0.777**	**0.663**	0.630	**0.570**	**0.670**	$\mathbf{0.631 \pm 0.075}$

- and on medical datasets (anomaly in elderly daily activities)

	Fall		Weakness		Nocturia		SlowerWalking		MoreTimeInChair			
	g-measure	F1	g-measure	F1	g-measure	F1	g-measure	F1	g-measure	F1	g-measure ($\mu \pm \sigma$)	F1 ($\mu \pm \sigma$)
LstmAE	0.085	0.014	0.182	0.108	0.000	0.000	0.158	0.049	0.133	0.035	0.112 ± 0.064	0.041 ± 0.037
AE	0.139	0.127	0.033	0.027	0.116	0.103	0.000	0.000	0.158	0.049	0.089 ± 0.062	0.061 ± 0.047
ConvAE	0.086	0.014	0.284	0.150	0.251	0.119	0.158	0.048	0.134	0.035	0.183 ± 0.074	0.073 ± 0.052
TadGAN*	0.222	0.267	0.570	0.555	0.000	0.000	**0.630**	**0.570**	0.267	0.222	0.338 ± 0.233	0.323 ± 0.216
HypAD (proposed)	**0.447**	**0.333**	**0.660**	**0.610**	**0.447**	**0.333**	0.470	0.364	**0.577**	**0.5**	$\mathbf{0.520 \pm 0.095}$	$\mathbf{0.428 \pm 0.123}$

- Geiger et al. (2020). “TADGAN: Time series anomaly detection using generative adversarial networks”. In IEEE Int. Conf. on Big Data’20.
- Diane J Cook, Aaron S Crandall, Brian L Thomas, and Narayanan C Krishnan. Casas: A smart home in a box. Computer, 46(7):62–69, 2012.

Hyperbolic Uncertainty for (Self-Paced) Self-Supervised Learning

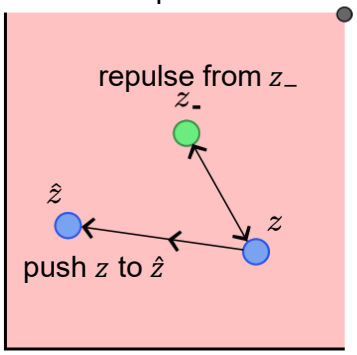
Franco, **Mandica**, Munjal, Galasso (2023). Hyperbolic Self-paced Learning for Self-supervised Skeleton-based Action Representations. In Proc. *ICLR*

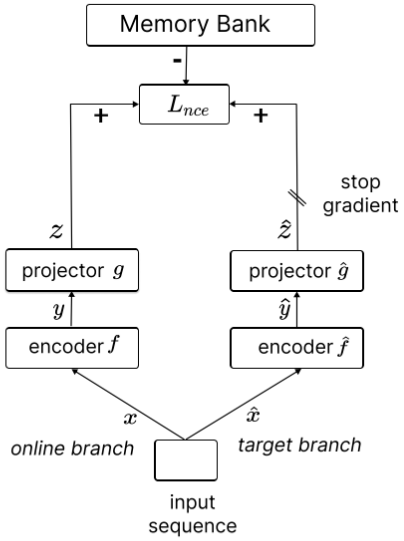


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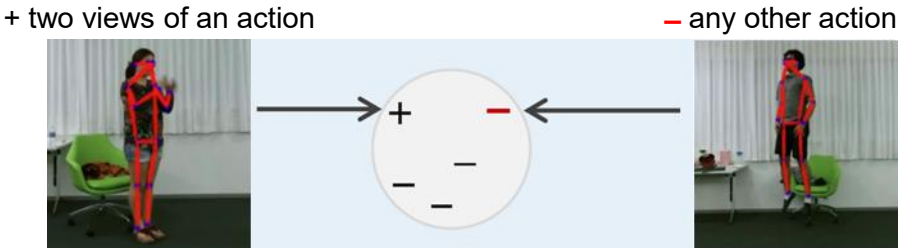
Skeleton-based SSL for action representations

- SOTA builds on SkeletonCLR [Li et al. CVPR'21]



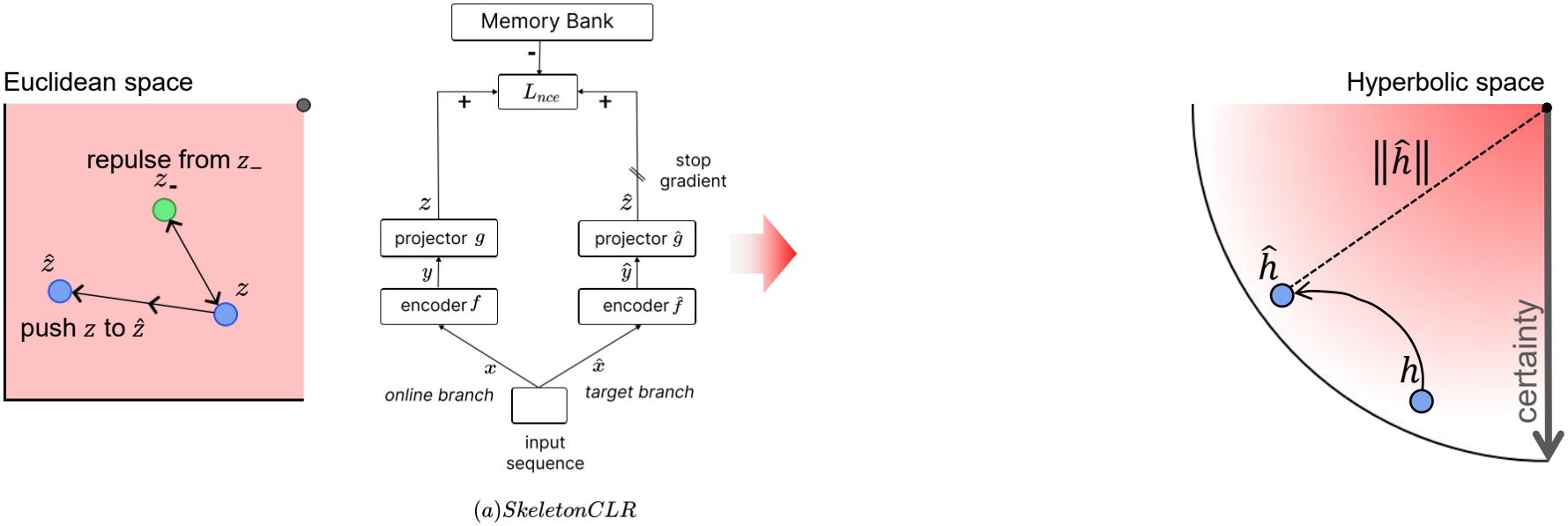


$(a) SkeletonCLR$



Hyperbolic Self-paced SSL (HYSP) ICLR'23

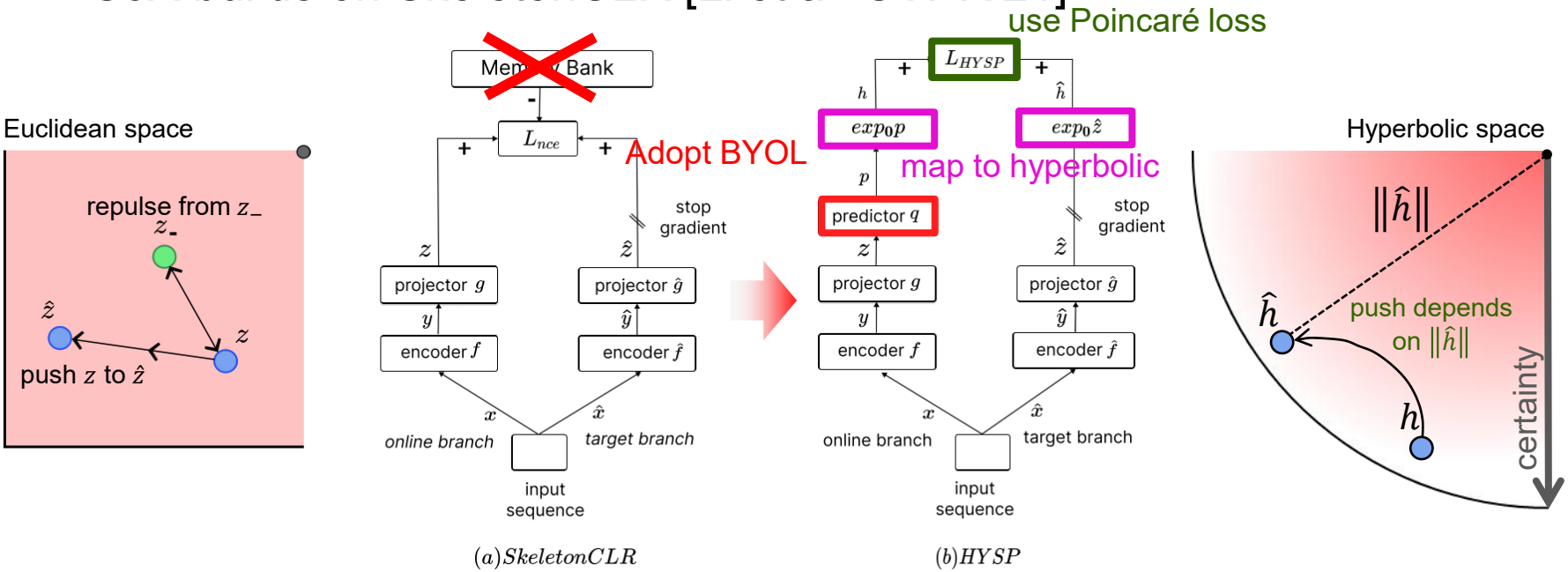
- SoA builds on SkeletonCLR [Li et al. CVPR'21]



- Proposition: use hyperbolic uncertainty to self-pace SSL (*HYSP*)
  - More certain samples should drive learning more predominantly

Hyperbolic Self-paced SSL (HYSP)
ICLR'23

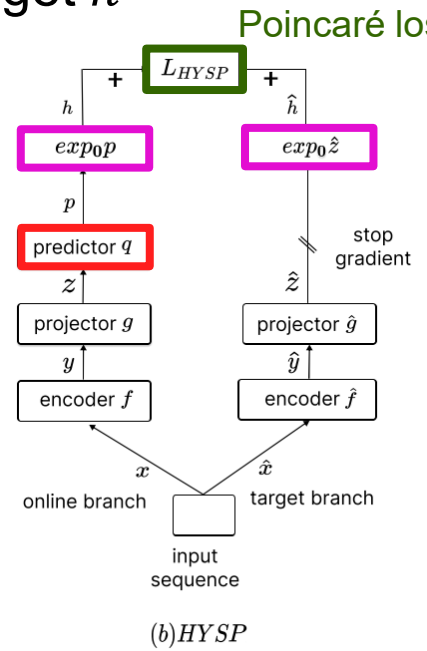
- SoA builds on SkeletonCLR [Li et al. CVPR'21]

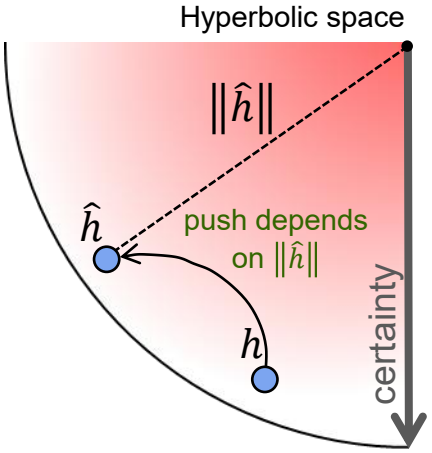


- Proposition: use hyperbolic uncertainty to self-pace SSL
  - More certain samples should drive learning more predominantly
- Hyperbolic Self-paced Self-Supervised Learning (HYSP)

Hyperbolic Self-paced SSL (HYSP) ICLR'23

- Learning: online h to match the target $\hat{h}$
- HYSP learning is self-paced
  - Poincaré loss gradient changes according to the certainty of the target $\hat{h}$
  - The larger the radius of $\hat{h}$, the more certain $\hat{h}$, the stronger are the gradients
  - More certain $\hat{h}$ drive learning more





$$L_{poin}(h,\hat{h}) = \cosh^{-1}\left(1 + 2\frac{\|h-\hat{h}\|^2}{(1-\|h\|^2)(1-\|\hat{h}\|^2)}\right)$$

$$\nabla L_{poin}(h,\hat{h}) = \frac{(1-\|h\|^2)^2}{2\sqrt{(1-\|h\|^2)(1-\|\hat{h}\|^2)+\|h-\hat{h}\|^2}}\left(\frac{h-\hat{h}}{\|h-\hat{h}\|} + \frac{h\|h-\hat{h}\|}{1-\|h\|^2}\right)$$

HYSP
Comparison with the state-of-the-art (SoA)

- Best on almost all SSL protocols on three established benchmarks

Datasets:

NTU60, NTU120, PKUMMD

Evaluation (NTU60):

- **Linear Protocol** **(+1.4%)**
- **Semi-supervised** **(+3.8%)**
- **Fine-Tuning** **(+2.4%)**

Results of linear, semi-supervised and finetuning protocols on NTU-60 and NTU-120

	Linear eval.				Semi-sup. (10%)		Finetune				Additional Techniques				
	NTU-60		NTU-120		NTU-60		NTU-60		NTU-120				Extra	Extra	
Method	*xsub*	*xview*	*xsub*	*xset*	*xsub*	*xview*	*xsub*	*xview*	*xsub*	*xset*	*3s*	*Neg.*	*Aug.*	*Pos.*	*ME*
P&C *Su et al. (2020)*	50.7	76.3	42.7	41.7	-	-	-	-	-	-					
MS^{2}L *Lin et al. (2020b)*	52.6	-	-	-	65.2	-	78.6	-	-	-		✓			
AS-CAL *Rao et al. (2021)*	58.5	64.8	48.6	49.2	-	-	-	-	-	-		✓			
SkeletonCLR *Li et al. (2021)*	68.3	76.4	56.8	55.9	66.9	67.6	80.5	90.3	75.4	75.9		✓			
MCC[§] *Su et al. (2021)*	-	-	-	-	55.6	59.9	83.0	89.7	79.4	80.8		✓			
AimCLR *Guo et al. (2022a)*	74.3	79.7	63.4	63.4	-	-	-	-	-	-		✓	✓	✓	
ISC *Thoker et al. (2021)*	76.3	**85.2**	**67.9**	**67.1**	65.9	72.5	-	-	-	-		✓	✓		✓
HYSP *(ours)*	**78.2**	82.6	61.8	64.6	**76.2**	**80.4**	**86.5**	**93.5**	**81.4**	**82.0**			✓		
3s-ST-GCN	-	-	-	-	-	-	85.2	91.4	77.2	77.1	✓				
3s-SkeletonCLR *Li et al. (2021)*	75.0	79.8	-	-	-	-	-	-	-	-	✓	✓			
3s-Colorization *Yang et al. (2021)*	75.2	83.1	-	-	71.7	78.9	88.0	94.9	-	-	✓				
3s-CrosSCLR *Li et al. (2021)*	77.8	83.4	67.9	66.7	74.4	77.8	86.2	92.5	80.5	80.4	✓	✓		✓	
3s-AimCLR *Guo et al. (2022a)*	78.9	83.8	**68.2**	**68.8**	78.2	81.6	86.9	92.8	80.1	80.9	✓	✓	✓	✓	
3s-HYSP *(ours)*	**79.1**	**85.2**	64.5	67.3	**80.5**	**85.4**	**89.1**	**95.2**	**84.5**	**86.3**	✓		✓		

Results of linear, semi-supervised and finetuning protocols on PKU-MMD I

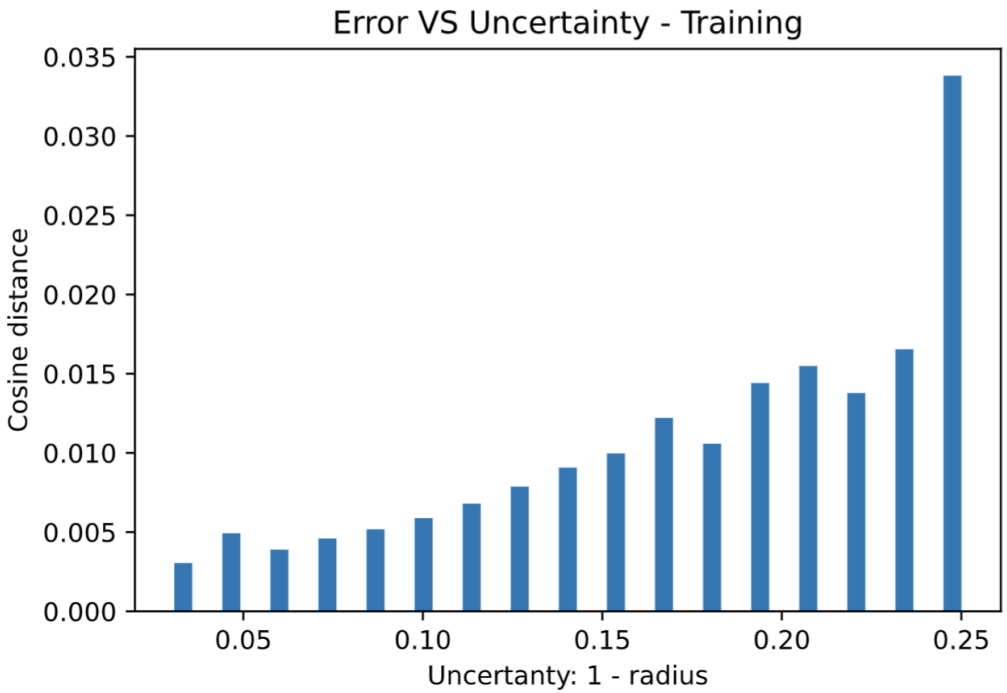
	Linear eval.				Semi-sup. (10%)				Finetune			
Method	*Joint*	*Bone*	*Motion*	*3s*	*Joint*	*Bone*	*Motion*	*3s*	*Joint*	*Bone*	*Motion*	*3s*
MS^{2}L *Lin et al. (2020b)*	64.9	-	-	-	70.3	-	-	-	85.2	-	-	-
SkeletonCLR *Li et al. (2021)*	80.9	72.6	63.4	85.3	-	-	-	-	-	-	-	-
ISC *Thoker et al. (2021)*	80.9	-	-	-	72.1	-	-	-	-	-	-	-
AimCLR *Guo et al. (2022a)*	83.4	82.0	**72.0**	87.8	-	-	-	86.1	-	-	-	-
HYSP *(ours)*	**83.8**	**87.2**	70.5	**88.8**	**85.0**	**87.0**	**77.8**	**88.7**	**94.0**	**94.9**	**91.2**	**96.2**

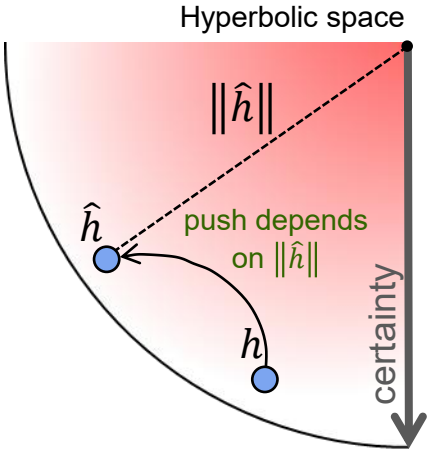
Results of transfer learning on PKU-MMD II

	Transfer Learning (PKU-MMD II)	
Method	*PKU-MMD I*	*NTU-60*
S+P *Zheng et al. (2018)*	43.6	44.8
MS^{2}L *Lin et al. (2020b)*	44.1	45.8
ISC *Thoker et al. (2021)*	45.1	45.9
HYSP *(ours)*	**50.7**	**46.3**

HYSP after training

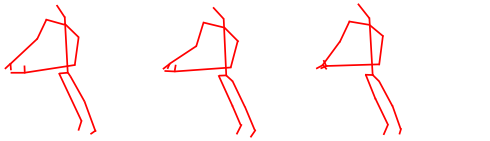
- End-to-end trained uncertainty matches the intuition
  - Large sample uncertainty corresponds to larger prediction errors (larger cosine distance)
  - Learn larger uncertainty for more ambiguous actions





HYSP - Qualitative results

Inter-Class Variability:
small movements, ambiguous movements, **larger unambiguous movements**



Playing with phone (Radius 0.7679)



Take off hat (Radius 0.8143)



Pushing (Radius 0.9697)

Intra-Class Variability: larger unambiguous motions get larger radii



Staggering (Radius 0.7725)



Staggering (Radius 0.873)



Staggering (Radius 0.9754)

Overview

- What is hyperbolic geometry?
- Overview of the field
- From Euclid to Hyperbolic Deep Learning
- Leading Interpretations of the Hyperbolic Radius
- Hyperbolic Uncertainty for Anomaly Detection
- Hyperbolic Uncertainty for (Self-Paced) Self-Supervised Learning
- Open Research Perspectives on the Hyperbolic Radius
- Closing remarks

Open Research Perspectives on the Hyperbolic Radius

Franco, Mandica, Kallidromitis, Guillory, Li, Darrell, Galasso (2023). Hyperbolic Active Learning for Semantic Segmentation under Domain Shift. ArXiv:2306.11180 pre-print

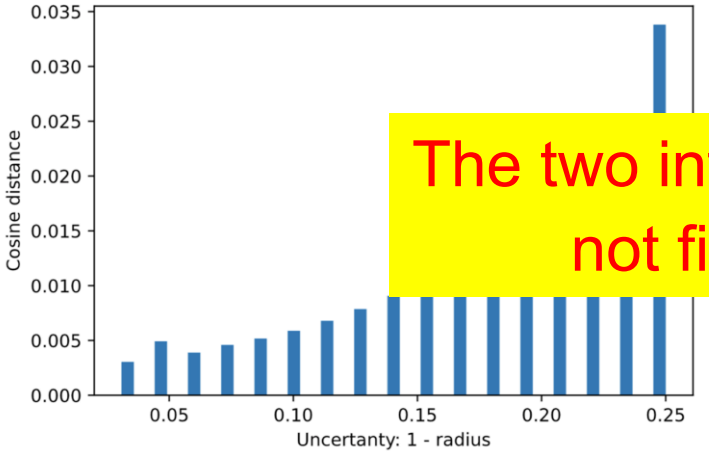


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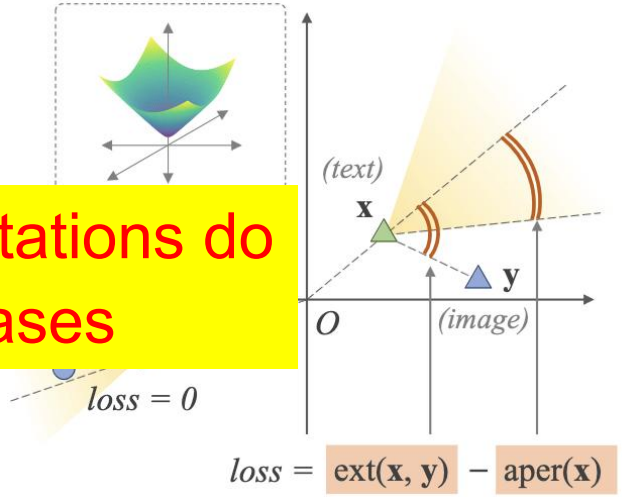
Recall the Leading Interpretations of the Hyperbolic radius

- Radius → certainty
  - Weight samples in SSL via the hyperbolic radius
    - **Franco et al. ICLR'23**
  - Detect anomalies and abstain if uncertain
    - **Flaborea et al. CVPR'23 Wks**
  - Larger error when more uncertain

- Radius → hierarchy (parent-to-child)
  - In hierarchical classification, parent classes have lower radii
    - Ghadimi Atigh et al. CVPR'22
  - Enforce image-text hierarchies by an entailment loss
    - Desai et al. PMLR'23



The two interpretations do not fit all cases



Open Research Perspectives on the Hyperbolic Radius

Franco, Mandica, Kallidromitis, Guillory, Li, Darrell, Galasso (2023).
Hyperbolic Active Learning for Semantic Segmentation under Domain Shift. ICML'24

D'Amely, Flaborea, Mettes, Galasso (2024).
Hyp^{2}Nav: Hyperbolic Planning and Curiosity for Crowd Navigation.
IROS 2024

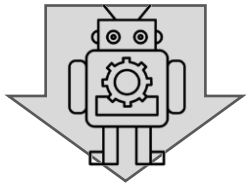


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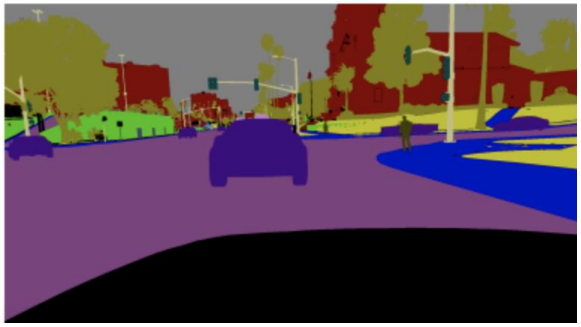
Active Domain Adaptation for Semantic Segmentation

GTA V (synthetic source dataset)





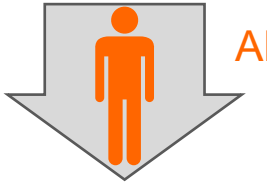
Automatically annotate all pixels



Domain adapt

Cityscapes (real target dataset)



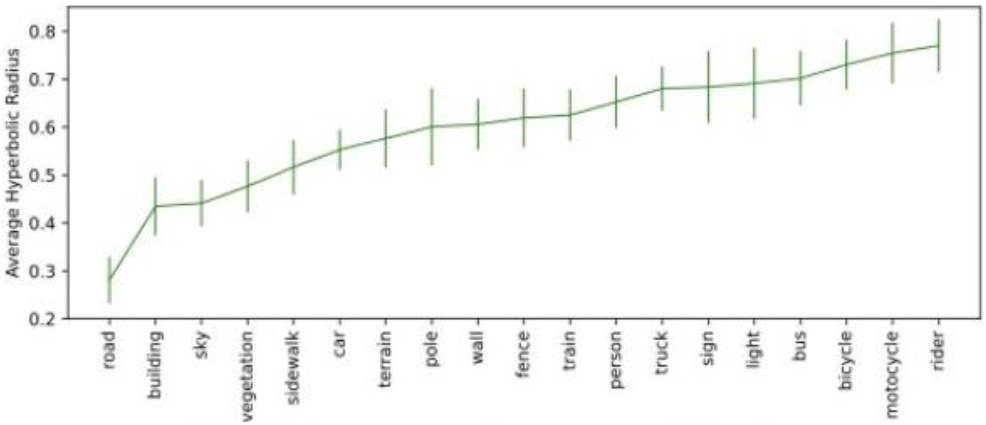


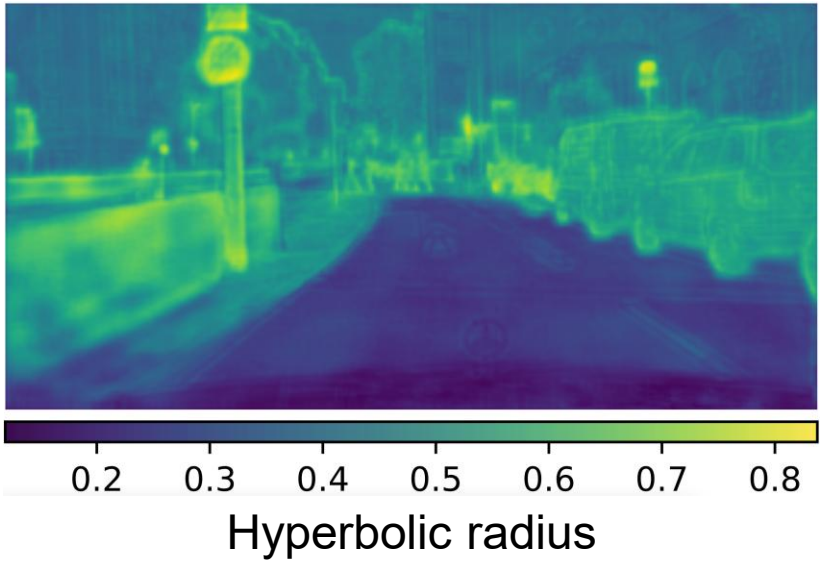
AL→ Manually label a few pixels in labelling rounds



Surprising hyperbolic radius

- Learn a SOTA semantic segmenter w/o enforcing hierarchies
  - Learned pixel embeddings via hyperbolic multinomial logistic regression

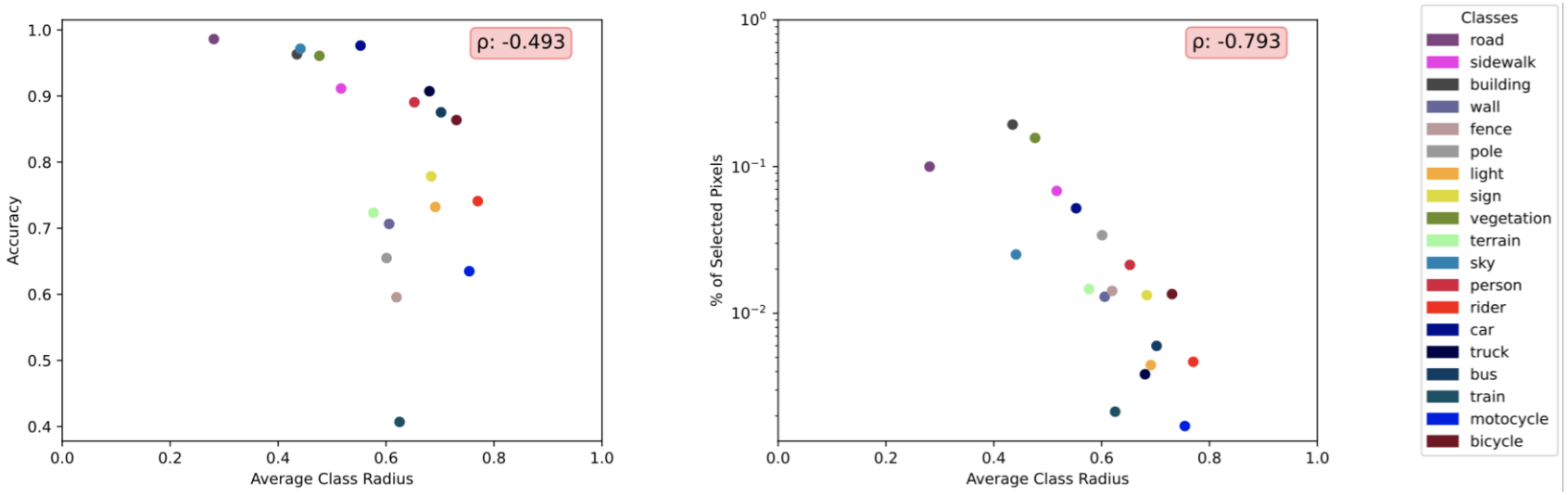




1. An object hierarchy do not emerge
2. The radius does not correspond to the pixel/class uncertainty
3. The model associates characteristic radii to each class
   → **classes are compact**

Bottom-up from data statistics and a novel interpretation of hyperbolic radius

- The average class radius negatively correlates with accuracy
  - Larger hyperbolic radius → larger error
- The average class radius negatively correlates with the number of selected pixel
  - More labelled pixels → lower hyperbolic radius



- Radius → class complexity (intrinsic complexity + data scarcity)

Open Research Perspectives on the Hyperbolic Radius

Franco, Mandica, Kallidromitis, Guillory, Li, Darrell, Galasso (2023). Hyperbolic Active Learning for Semantic Segmentation under Domain Shift. ArXiv:2306.11180 pre-print

D'Amely, Flaborea, Mettes, Galasso (2024).
Hyp^{2}Nav: Hyperbolic Planning and Curiosity for Crowd Navigation.
IROS 2024



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Motivations

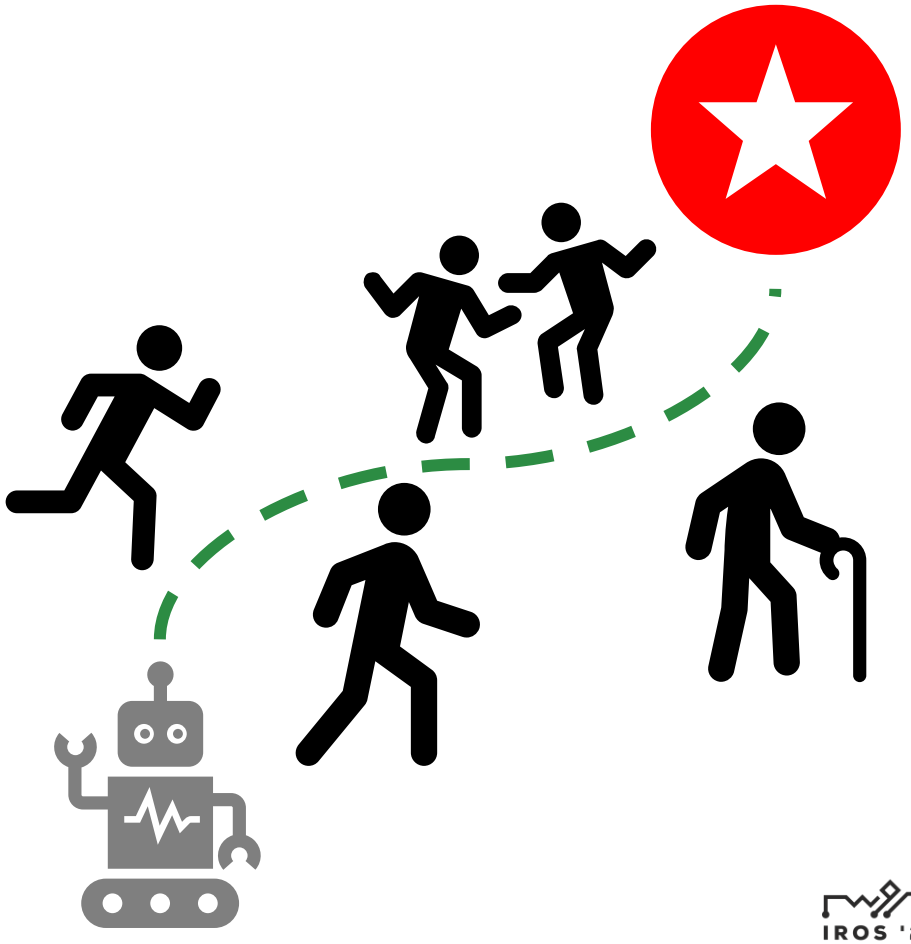
Crowd Navigation

Fast

the agent has to reach the goal in the minimum travel time

Safe

the agent should calibrate its path dynamically, considering the moving obstacles occurring on its trajectory



Motivations

Fast

the agent has to reach the goal in the minimum travel time

Safe

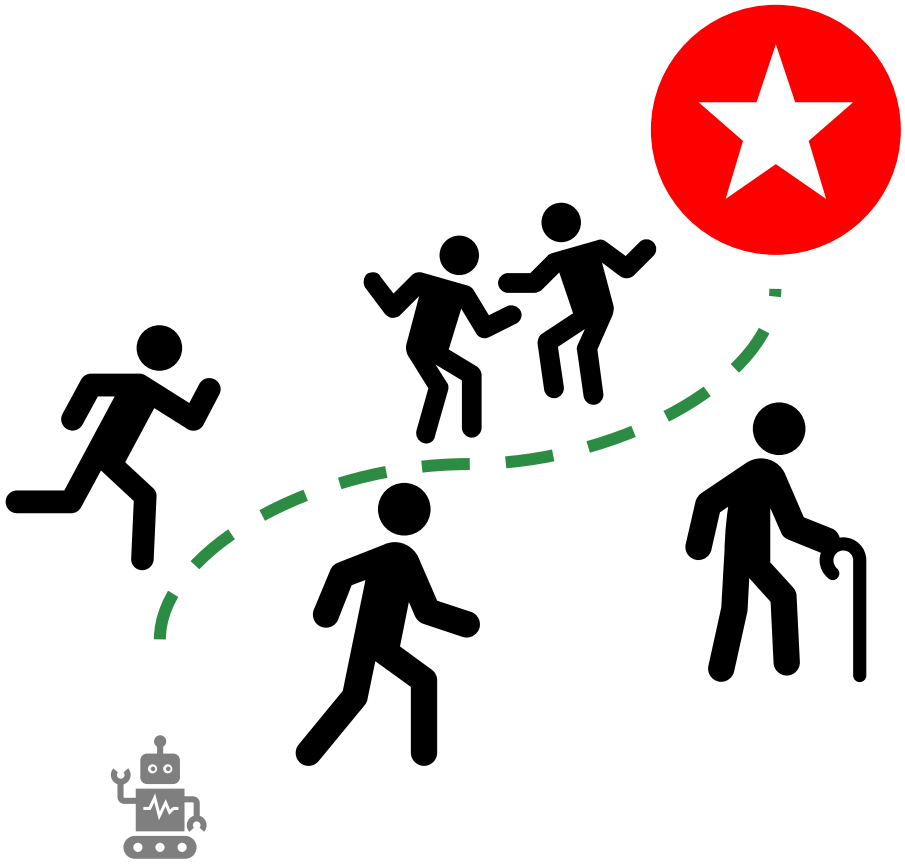
the agent should calibrate its path dynamically, considering the moving obstacles occurring on its trajectory

Computational constraints

Current systems use high-dimensional representation which can result in a bottleneck in computation

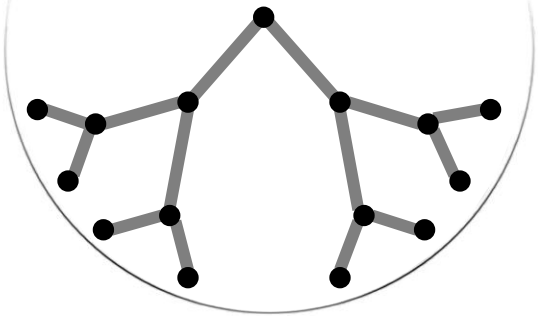
Unpredictability of human behavior

Resulting in failures of the robot policy, demanding for greater interpretability



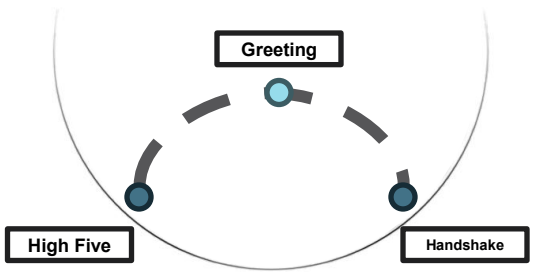
IROS '24 ABU DHABI

Hyperbolic Learning

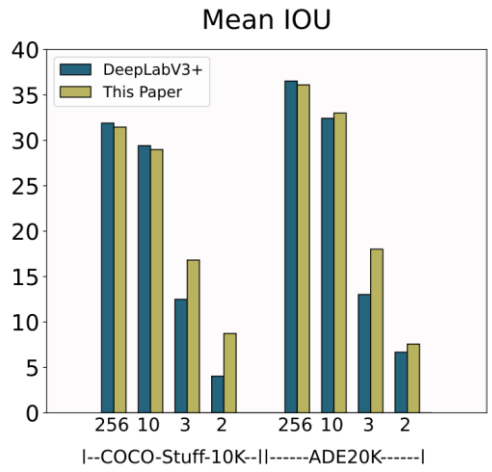


Convenient Embedding Space for Graphs





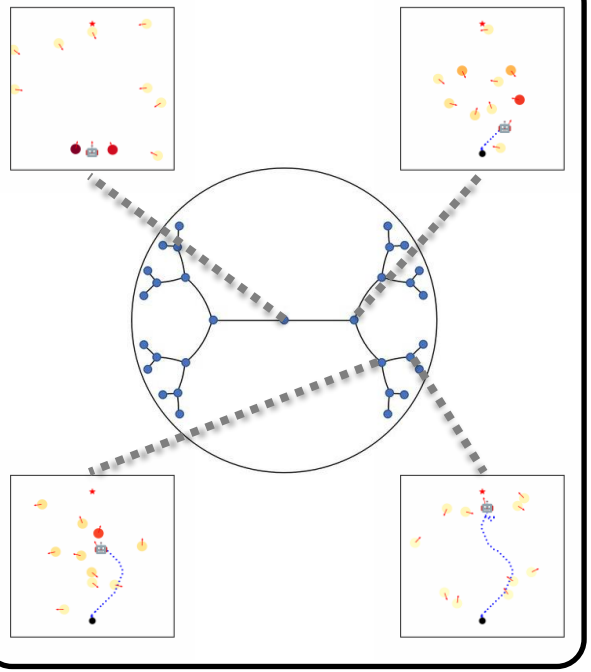
Improved Interpretability



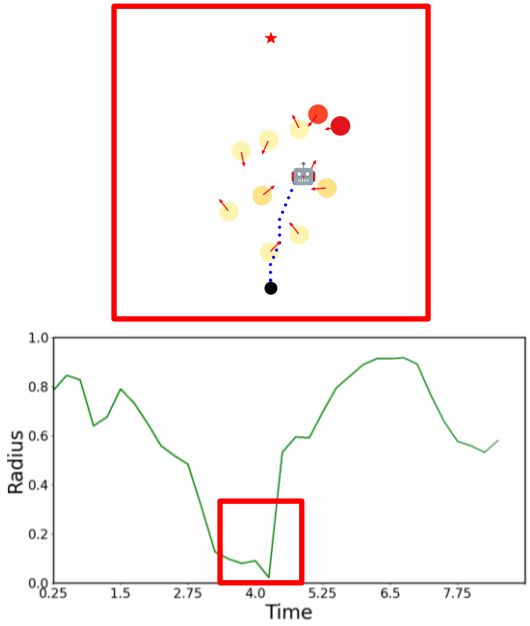
Effective Low-dimensional Embeddings

[1]: E. Cetin, B. P. Chamberlain, M. M. Bronstein, and J. J. Hunt, "*Hyperbolic deep reinforcement learning*," in The Eleventh International Conference on Learning Representations, 2023.
[2]: D. Surís, R. Liu, and C. Vondrick. "*Learning the predictability of the future*". In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021.
[3]: M. G. Atigh, J. Schoep, E. Acar, N. Van Noord, and P. Mettes, "*Hyperbolic image segmentation*". In IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022.

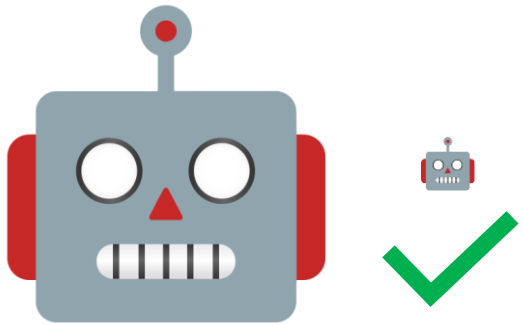
Hyperbolic Learning for Crowd Navigation



Convenient Embedding Space for Graphs (as MDPs)

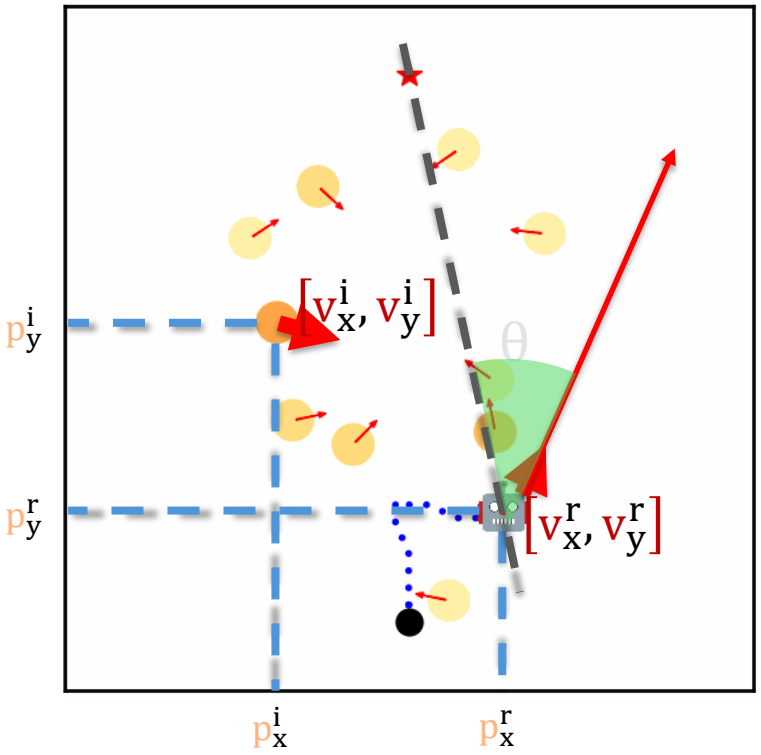


Improved Interpretability

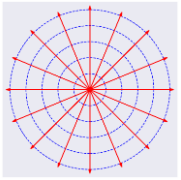


Low-dimensional Embeddings

Problem Formulation



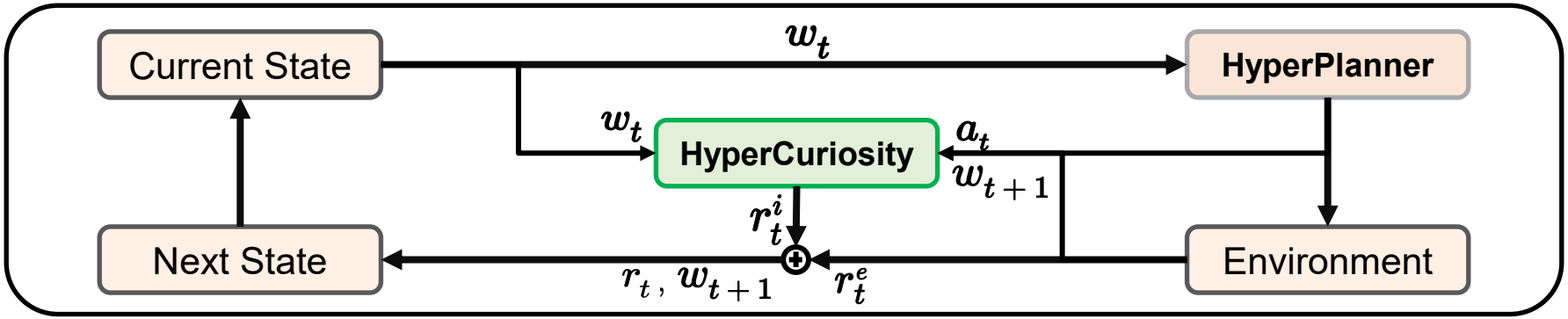
- Human State:
$$s(h_i) = [p_x^i, p_y^i, v_x^i, v_y^i, r]$$
- Robot State:
$$s(r) = [p_x^r, p_y^r, v_x^r, v_y^r, \theta, v_M, r]$$
- Actions:
  - 16 directions
  - 5 velocities
  - [STOP]



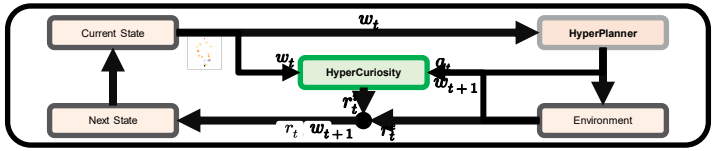
- Experiments:
  - **Simple** Scenario: Train: 5 Obstacles, Test: 5 Obstacles
  - **Complex** Scenraio: Train: 10 Obstacles, Test: 10 Obstacles
  - **Complex+** Scenario: Train: 10 Obstacles, Test: 20 Obstacles

[4]: C. Chen, Y. Liu, S. Kreiss, and A. Alahi, “Crowd-robot interaction: Crowd-aware robot navigation with attention-based deep reinforcement learning”. In International Conference on Robotics and Automation, 2019.

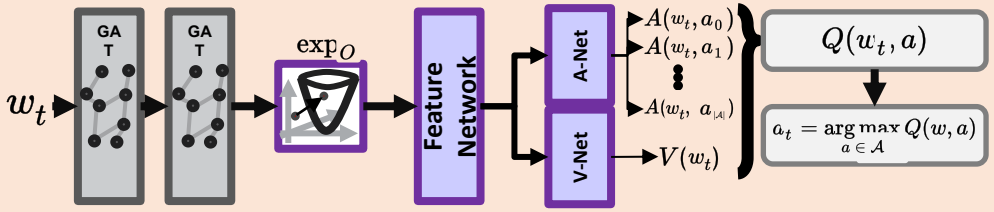
Hyp^{2}Nav


Current State
w_t
HyperPlanner
w_t
HyperCuriosity
a_t
w_{t+1}
r_t^i
Next State
Environment
r_t , w_{t+1}
r_t^e

Hyp²Nav

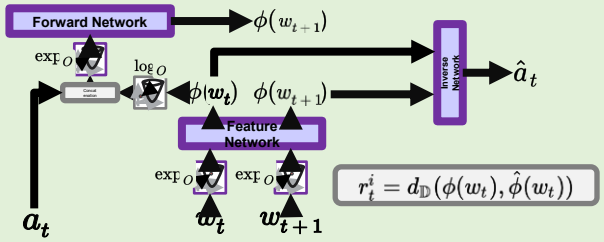


HyperPlanner



- Leverages exponential mapping to transform representations into hyperbolic space and h-MLP for feature extraction and refinement
- Leverages hyperbolic geometry to maintain expressive power despite low dimensionality, achieving efficient computation
- Leverages hierarchical structure encoding capabilities of hyperbolic space

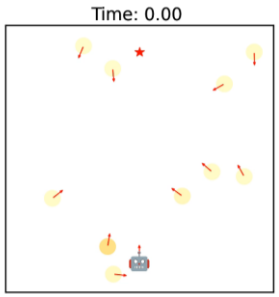
HyperCuriosity

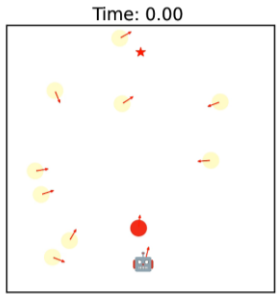


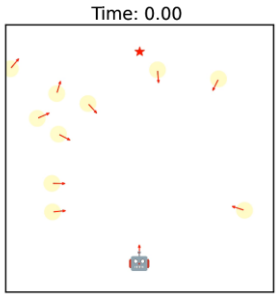
- Refines exploration-exploitation trade-off using hyperbolic space properties
- Extends Intrinsic Curiosity Module (ICM)
- Computes intrinsic rewards based on discrepancy between actual and predicted next-state representations in hyperbolic space

Qualitative Comparison

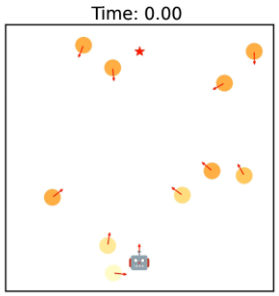
SafeCrowdNav

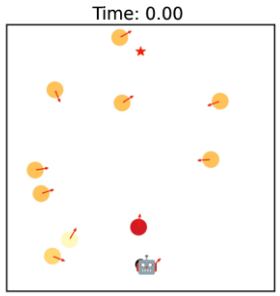


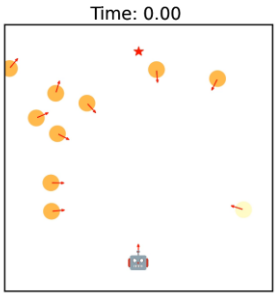




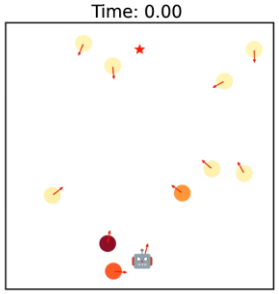
Intrinsic-SGD3QN

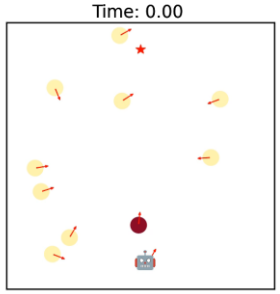


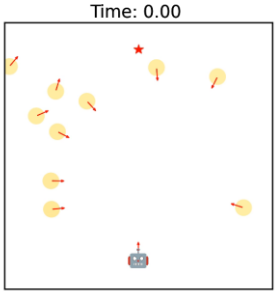




Hyp^{2}Nav
(proposed)

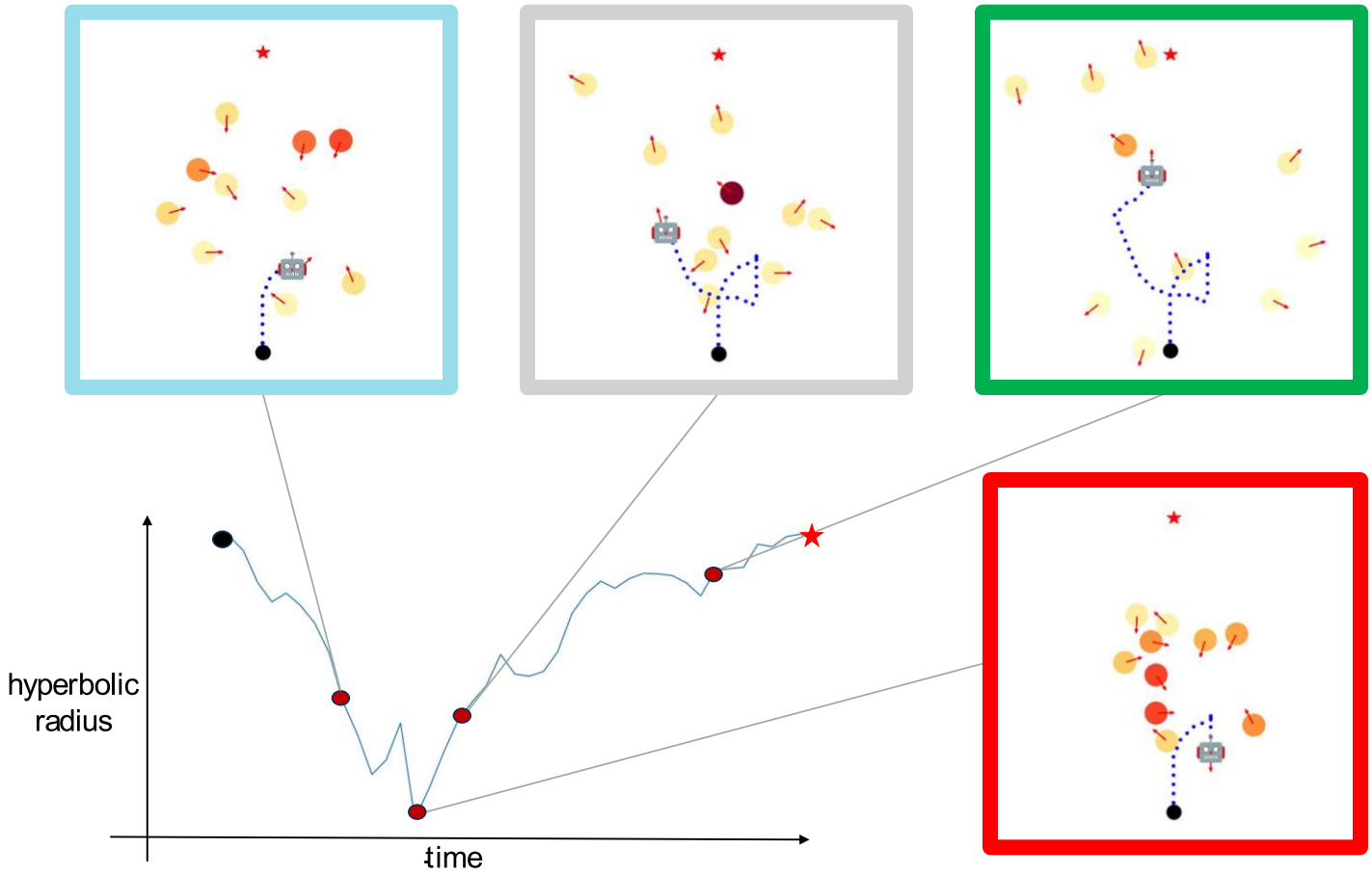






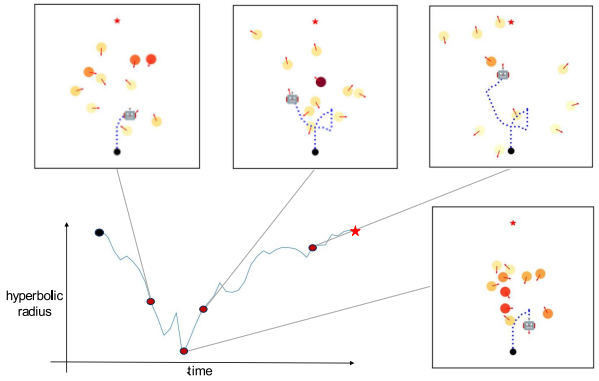
IROS '24
ABU DHABI

Gauging the agent uncertainty

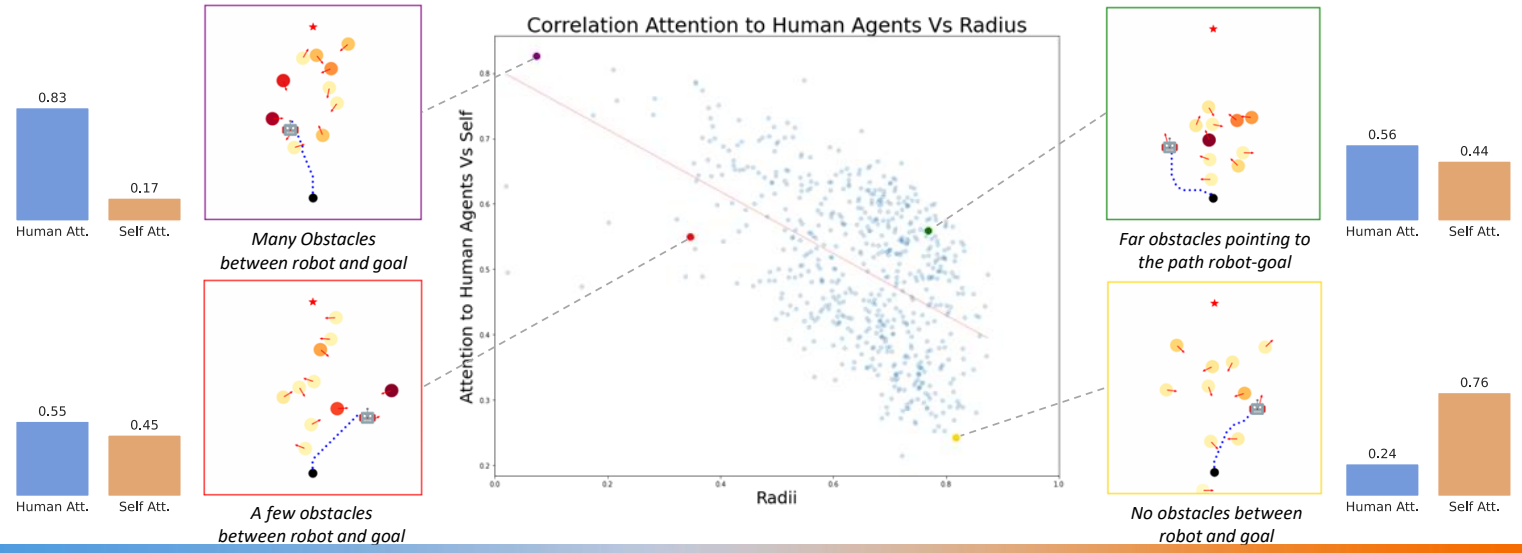


IROS '24
ABU DHABI

Gauging the agent uncertainty

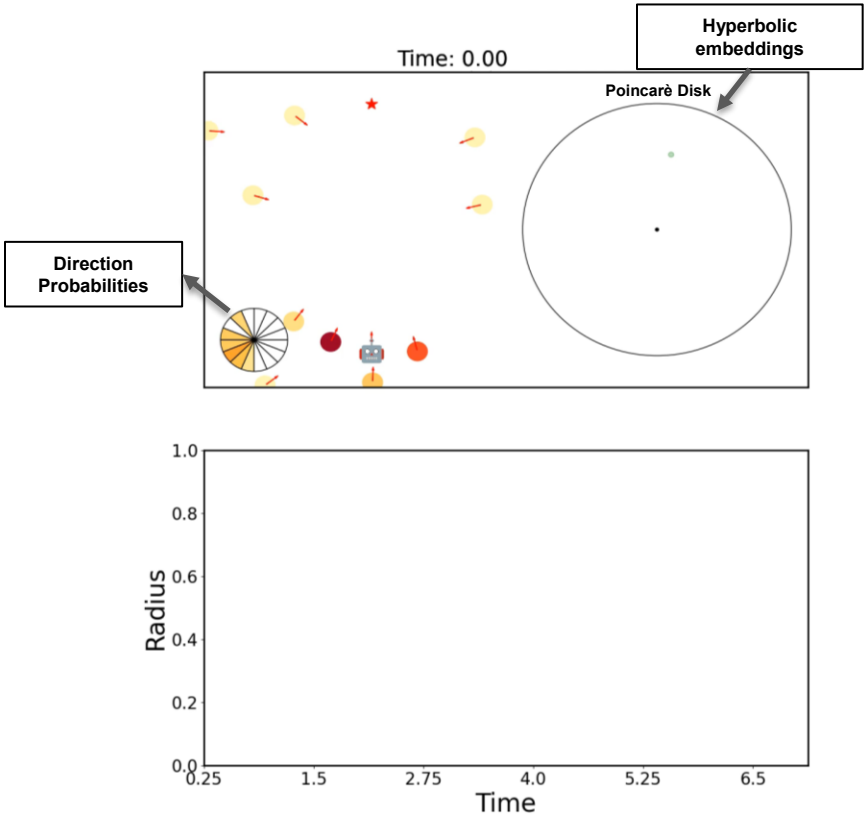


- Hyperbolic radius as an interpretable measure of the agent's certainty and situation complexity
- **Smaller radius = more complex situations.** Increased attention to possible collisions, reflecting increased uncertainty in the agent's decision-making
- **Larger = easier situations.** The path connecting the agent to the goal is clear, showing the agent's high confidence
- The model correctly predict obstacle intentions and anticipate future collision risks

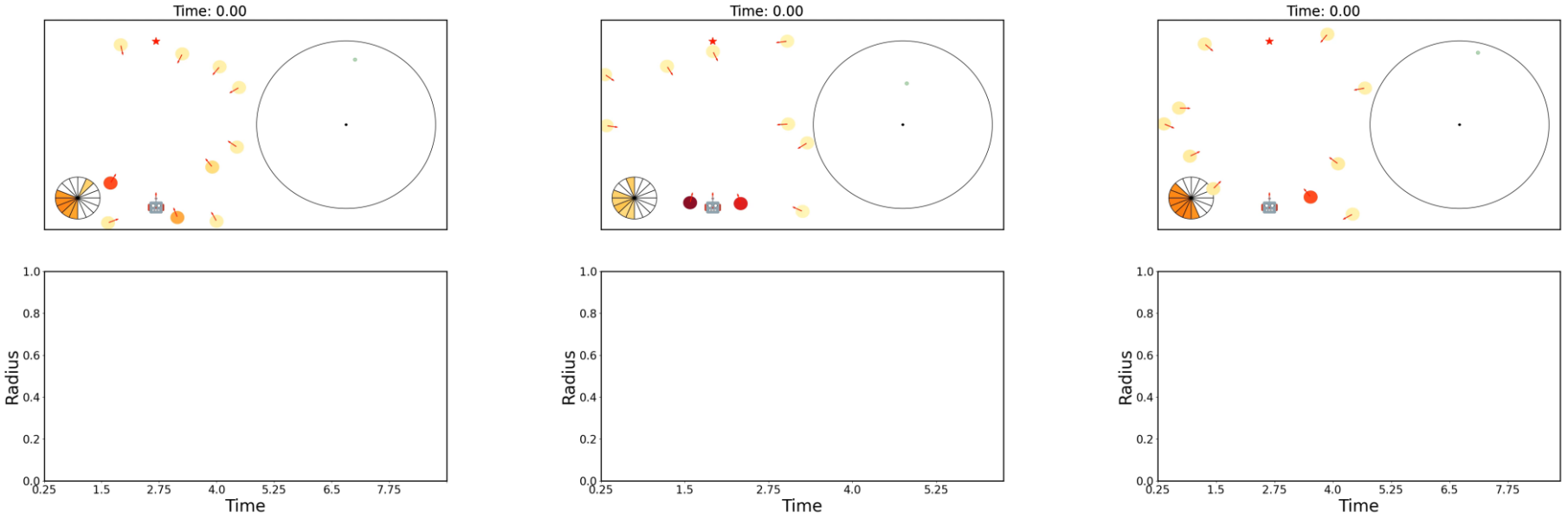


IROS '24 ABU DHABI

Qualitative Comparison



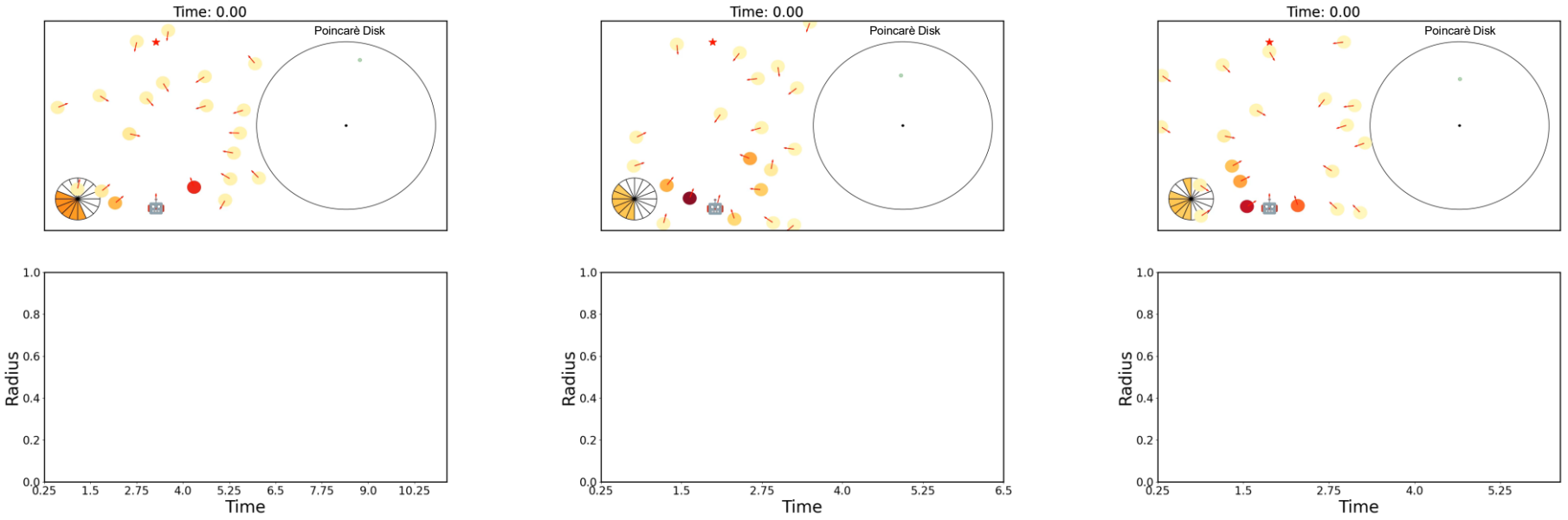
Radius Uncertainty - **Complex** Scene



Results obtained with Hyp^{2}Nav

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Radius Uncertainty - **Complex +** Scene



Results obtained with Hyp^{2}Nav

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Overview

- What is hyperbolic geometry?
- Overview of the field
- From Euclid to Hyperbolic Deep Learning
  - Riemmanian optimizers
  - Hyperbolic radius
  - Entailment cones
- Leading Interpretations of the Hyperbolic Radius
  - Hyperbolic Uncertainty for Anomaly Detection
  - Hyperbolic Uncertainty for (Self-Paced) Self-Supervised Learning
- Open Research Perspectives on the Hyperbolic Radius
- Closing remarks

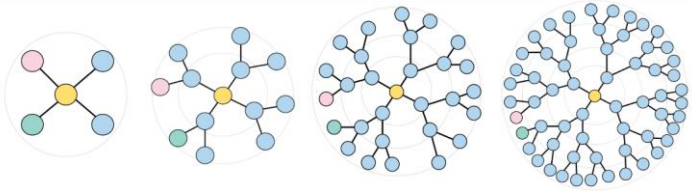
Closing remarks



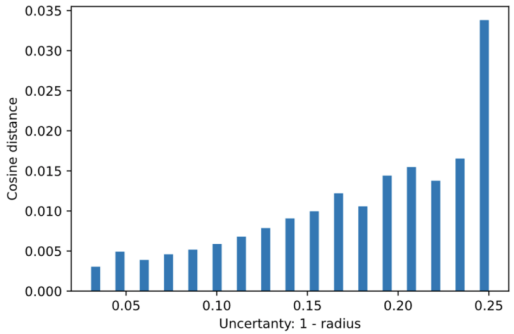
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Hyperbolic learning: a new field in computer vision

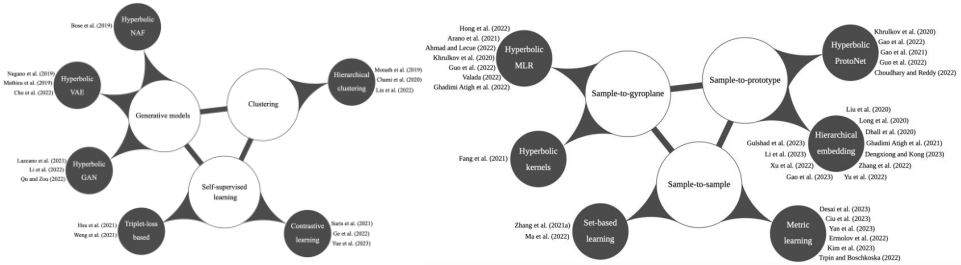
- Visual data and labels are commonly hierarchical
  - Euclidean space distorts hierarchies
  - Hyperbolic space is compact and a natural hierarchical geometry



- Uncertainty estimation is a long-standing goal of ML
  - Hyperbolic geometry allows end-to-end estimation of uncertainty



- Exponential paper growth at top-tier CV and ML conferences and several open theoretic and experimental challenges



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Further material and acknowledgements

- Our open-source code for HypAD and HYSP available at:
  - https://github.com/paolomandica/hysp
  - https://github.com/aleflabo/HypAD
- Our code for HALO (https://arxiv.org/abs/2306.11180) will be available soon on https://www.pinlab.org/
- Slide credits to Pascal Mettes, Max van Spengler, Yunhui Guo, and Stella Yu and to their tutorial on Hyperbolic Deep Learning for Computer Vision at CVPR'23 (https://sites.google.com/view/hdlcv-cvpr23tutorial)
- Recommended further reading material:
  - Mettes et al. (2023). Hyperbolic Deep Learning in Computer Vision: A Survey. Pre-print arXiv:2305.06611.
  - Ganea Becigneul Hofmann NeurIPS 2018 Hyperbolic neural networks
  - Shimizu Mukuta Harada ICLR 2021 Hyperbolic Neural Networks++
  - Blog post by Keng: https://bjlkeng.io/posts/hyperbolic-geometry-and-poincare-embeddings/

Thank you

Acknowledgments: Pascal Mettes, Guido D'Amely



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